

# PPO, GRPO, and On-Policy Distillation for Long-Horizon Post-Training

Limes Labs Research Workstream

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## Abstract

Critic-free reinforcement learning, especially Group Relative Policy Optimization (GRPO), is attractive for reasoning-model post-training because it removes the learned value model and reduces memory pressure. This paper asks where that trade changes in long-horizon, variable-length, tool-using trajectories. Our working hypothesis is conditional: critic-based PPO is a better candidate when token- or state-level temporal credit is observable and covered, while GRPO remains compelling when group reward contrast is reliable and value modeling is weak or too expensive. OPD and OPSD are treated as complementary on-policy distillation objectives, not direct replacements for policy optimization. We formalize the method taxonomy, cost accounting, and failure modes, and run a CPU-only toy benchmark with known oracle advantages. Across a 20-seed, 18-case matrix, critic-style temporal-difference advantages have higher mean oracle-advantage correlation in 17 cases; a confidence-interval reading gives 16 clear critic-favorable cases, 1 near tie, and 1 clear group-favorable counterexample. The result is mechanism evidence, not a frontier-model leaderboard claim.

**Keywords.** post-training; reinforcement learning from verifiable rewards; PPO; GRPO; on-policy distillation; credit assignment; long-horizon agents.

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# 1 Introduction

Long-horizon post-training for language-model agents is not merely long-context language modeling. A trajectory may include partial plans, tool calls, retrievals, edits, failed attempts, retries, compacted summaries, and final verifier feedback. In such traces, the question “did the rollout succeed?” is not the same as “which token or action helped?” A scalar terminal reward can be useful for ranking responses, but it can also broadcast the same update over helpful subgoals, harmful detours, and no-op padding.

This paper is a public Limes Labs workstream on the method question: *for variable-length, multi-step, tool-heavy trajectories, when does a critic/value-model setup recover useful temporal credit that a group-relative setup discards or blurs?* We deliberately avoid centering the work on one base model. Z.ai’s GLM-5.2 report is discussed only as an industry case study: it describes compacted, variable-length long-horizon rollouts and a move from group-wise optimization to critic-based PPO with token-level advantages [14]. That is not independent causal evidence that PPO beats GRPO; it is a motivating example of the problem structure studied here.

**Contributions.** We make four contributions. First, we separate PPO, GRPO, OPD, and OPSD by objective, signal granularity, and extra-model cost. Second, we give a cost and failure-mode checklist for fair post-training comparisons. Third, we introduce a small deterministic toy environment with oracle advantages and compare group-relative and critic-style advantage estimators across a multi-seed matrix. Fourth, we state testable predictions for future neural and agentic experiments.

**Scope discipline.** The paper makes a narrow claim about credit-assignment mechanisms. It does not claim that PPO is universally better than GRPO, that OPD is weaker than RL, or that a finite-state toy predicts frontier-model behavior. We use the toy because the oracle advantage is known, which lets us ask a question that is usually hidden in language-model training: whether an estimator is aligned with the token-level causal structure of a trajectory. This distinction matters for public research. A method can improve a benchmark for reasons unrelated to the mechanism under discussion, and a method can be mechanistically promising while still losing under a particular compute budget, implementation, reward model, or data mixture.

**Paper organization.** Section 2 defines the methods and separates optimization from distillation. Section 3 reviews related work. Section 4 gives cost accounting. Sections 5–7 define the toy environment, experimental protocol, and results. Sections 8–10 interpret the mechanism evidence, compare method regimes, and state failure modes. The appendices provide implementation details, case design, full summary tables, and raw per-seed rows generated directly from the canonical JSON result artifact.

## 2 Definitions and Method Taxonomy

Let a policy produce a trajectory

$$\tau = (s_0, a_0, s_1, a_1, \dots, s_T), \tag{1}$$

where a state may include the prompt, partial response, tool outputs, verifier state, files, memory, or compacted context. Let  $R(\tau)$  be an outcome reward, possibly combined with per-token KL, process rewards, or unit-test feedback.

## 2.1 PPO with a critic

PPO optimizes a clipped policy-gradient surrogate [10]:

$$L^{\text{PPO}}(\theta) = \mathbb{E}_t [\min(\rho_t(\theta)A_t, \text{clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon)A_t)], \quad (2)$$

where

$$\rho_t(\theta) = \frac{\pi_\theta(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)}. \quad (3)$$

In actor-critic use, a value model  $V_\phi$  can estimate one-step temporal difference errors

$$\delta_t = \nabla_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t), \quad (4)$$

or generalized advantage estimates [9]:

$$A_t^{\text{GAE}(\gamma, \lambda)} = \sum_{\ell=0}^{\infty} (\gamma \lambda)^\ell \delta_{t+\ell}. \quad (5)$$

The critic is valuable only if the value target is learnable, sufficiently covered, and cheaper than the credit it recovers.

## 2.2 GRPO

GRPO samples a group of completions for the same prompt and normalizes rewards inside the group [2, 11]. A simplified response-level advantage is

$$\hat{A}_i^{\text{GRPO}} = \frac{R_i - \mu_G}{\sigma_G + \epsilon}, \quad (6)$$

where  $\mu_G$  and  $\sigma_G$  are group statistics. This critic-free design is memory-efficient and effective in many verifiable-reward settings. Its long-horizon risk is intra-rollout information loss: one scalar can be broadcast across tokens with very different causal roles.

## 2.3 OPD and OPSD

OPD trains a student on student-sampled trajectories while a teacher provides dense supervision on those visited states [1, 3]. A typical objective is

$$L^{\text{OPD}} = \mathbb{E}_{x \sim D, y \sim \pi_s(\cdot | x)} \left[ \sum_t D_{\text{KL}}(\pi_T(\cdot | x, y_{<t}) \parallel \pi_s(\cdot | x, y_{<t})) \right]. \quad (7)$$

OPSD removes the separate teacher model by making one model play a privileged teacher role and an ordinary student role over student rollouts [15]. These are dense distillation objectives. They can be excellent post-training tools, but they are not the same objective as reward optimization.

## 3 Related Work

**PPO and actor-critic post-training.** PPO was introduced as a simpler clipped-objective alternative to trust-region policy optimization [10]. In RLHF-style language-model training, actor-critic optimization is usually wrapped with KL penalties, reference policies, reward models, old-logprob storage, reward normalization, and sequence-level preference signals [7, 12]. The relevant point for this work is not one historical RLHF recipe, but the availability of a state- or token-indexed value estimate. A critic can turn delayed outcome feedback into temporally structured advantages. That can help when rollouts contain subgoals and detours, but it also creates a second model that can drift, overfit, or assign confident false credit.

Table 1: Method taxonomy for long-horizon post-training.

| Method       | Objective               | Signal granularity    | granu-Extra signal             | Main risk                         |
|--------------|-------------------------|-----------------------|--------------------------------|-----------------------------------|
| PPO + critic | Reward/KL optimization  | token/state           | value model, reward, reference | critic cost and miscalibration    |
| GRPO         | Group-normalized RL     | group/response scalar | reward or verifier, no critic  | weak intra-rollout credit         |
| OPD          | Teacher distillation    | token logits          | teacher model                  | teacher mismatch and teacher cost |
| OPSD         | Privileged distillation | self- token logits    | privileged traces              | privileged leakage and truncation |

**GRPO and RL with verifiable rewards.** DeepSeekMath introduced GRPO as a critic-free method for mathematical reasoning, replacing the learned value model with group-relative reward normalization [11]. DeepSeek-R1 then made GRPO-style RL a central public reference point for reasoning-model training [2]. Subsequent work has analyzed the effective loss, success amplification, response-length effects, dynamic sampling, clipping details, and statistical structure of GRPO-like gradients [4, 6, 8, 13, 16]. The practical attraction is real: GRPO removes critic memory and critic training, which can be decisive at large scale. The risk studied here is that a response-level group scalar can be information-wasteful inside long, heterogeneous rollouts.

**On-policy distillation.** OPD and related generalized knowledge distillation methods train on student-visited states, reducing train/inference mismatch relative to purely offline teacher forcing [1, 3]. OPSD uses the same model in privileged teacher and ordinary student roles, aiming to capture dense reasoning supervision without a separate teacher model [15]. Recent OPD analyses emphasize teacher/student compatibility, teacher novelty, length inflation, and truncation collapse [3, 5]. These methods are close to the present problem because they also produce dense token-level signals on rollouts. They differ because their signal is a distillation target rather than a reward-improvement objective.

**Long-horizon agentic training.** Long-horizon agentic RL introduces issues that are often absent in short verifier tasks: tool-call state, file-system state, hidden evaluation artifacts, environment side effects, compaction, and anti-hacking infrastructure. The GLM-5.2 report is useful because it explicitly names compacted sub-traces, variable numbers of trainable traces per prompt, and a move from group-wise optimization to critic-based PPO with token-level advantages [14]. We treat this as a case study of a design pressure, not as proof of a causal performance mechanism.

## 4 Cost Accounting

Fair comparisons should charge all hidden inputs, not only final benchmark score. For PPO, the bill includes value-model parameters, value targets, critic forward passes, optimizer state, old-policy log probabilities, KL control, and rollout storage. For GRPO, the bill includes group size, discarded completions, low-dispersion groups, and verifier calls. For OPD/OPSD, it includes teacher or privileged-context compute, dense logits, trajectory length inflation, and truncation handling. For agentic coding RL, it also includes anti-hacking filters, online tool-call monitors, LLM judges, sandboxing, blocked-call handling, and dummy observations when invalid calls are intercepted.

The near-term reporting unit for this workstream is:

$$\frac{\Delta \text{reward}}{\text{generated tokens} \times \text{wall clock} \times \text{extra model memory}}, \quad (8)$$

reported alongside reproducible artifacts and negative results.

**Policy and rollout cost.** For PPO, each trainable token typically needs old-policy log probabilities, advantages, reference-policy information for KL control, and value-model targets. If rollouts are compacted into sub-traces, the implementation must decide whether every sub-trace is trainable, how truncation is handled, and how losses are normalized across very different lengths. GRPO avoids the critic but spends compute on group sampling. If many group members have identical verifier outcomes, the effective signal can be weak even when token generation cost is high.

**Verifier, judge, and safety cost.** In coding-agent RL, pass/fail rewards can be corrupted by shortcuts such as reading hidden tests, fetching target solutions, or exploiting benchmark leakage. Online anti-hacking filters, rule-based monitors, LLM judges, dummy observations for blocked calls, and sandboxing are not optional bookkeeping details. They are part of the training objective as experienced by the agent. Ignoring them makes method comparisons look cleaner than they really are.

**Distillation cost.** OPD and OPSD can look cheap if only student updates are counted. A fair bill also includes teacher logits or hidden states, privileged traces, context construction, rejection/truncation policies, and any additional sampling needed to keep the student on-policy. Dense token supervision can be worth this cost, but it should not be compared to RL methods as if the dense signal were free.

## 5 Toy Experiment

### 5.1 Environment

The toy environment creates variable-length trajectories with three action types: **help**, **harm**, and **wait**. **help** increases a hidden progress score, **harm** decreases it, and **wait** consumes a token without changing progress. A terminal verifier returns success when the final score reaches a prompt-specific threshold. Because the finite dynamics are known, we can compute an oracle state-action advantage:

$$A^*(s_t, a_t) = -c + V^*(s_{t+1}, h - 1) - V^*(s_t, h), \quad (9)$$

where  $c$  is token cost and  $h$  is remaining horizon.

### 5.2 Estimators and metrics

The group-relative estimator normalizes terminal returns inside each prompt group and broadcasts the resulting scalar to all tokens in the trajectory. The critic-style estimator fits a tabular value model from sampled returns-to-go and computes one-step temporal-difference advantages. We measure Pearson correlation with the oracle advantage, calibrated MSE after affine rescaling, raw MSE, sign accuracy, wait-token leakage, within-trajectory variance, zero-variance group fraction, and exact critic-state hit rate.

This is not a closed-loop PPO-vs-GRPO training benchmark. It is an estimator fidelity experiment under controlled dynamics.

## 6 Experimental Protocol

**Design principle.** The experiment is designed around a known oracle. In real language-model RL, we usually observe a reward and a model update but not the true token-level advantage. Here the finite-state dynamics allow exact dynamic programming for  $V^*$ , which turns the question into an estimator-fidelity problem. That lets us separate two claims that are often conflated: whether a method is cheap and whether its advantage signal points at the right tokens.

**Case axes.** The deep matrix varies five axes. The horizon axis tests whether scalar terminal broadcasts become less aligned as trajectories lengthen. The group-size axis tests whether more completions per prompt rescue GRPO-like normalization. The critic-budget axis tests coverage: a value table trained from too few groups should not be trusted. The observability axis removes or coarsens state features to simulate poor value-model inputs. The sparse-reward axis tests harder terminal feedback. Finally, the counterexample intentionally combines a blind critic and low coverage to create a regime where group-relative terminal outcomes should win.

**Decision rule.** The headline statistic is Pearson correlation with the oracle advantage. A case is a mean critic win when the mean critic correlation exceeds the mean group-relative correlation. Because one mean win can be tiny, we also report a 95% across-seed confidence interval for the delta. A “clear critic” case has positive delta after subtracting the interval; a “clear group” case has negative delta after adding the interval; otherwise the case is labeled a near tie. This is why the abstract says both “17 mean critic wins” and the more cautious “16 clear, 1 near tie, 1 clear group” reading.

**Why not a closed-loop toy RL run?** A closed-loop toy PPO/GRPO implementation is a useful next step, but it would mix estimator fidelity with optimizer settings, clipping schedules, policy parameterization, exploration, and KL control. The current paper asks a more basic question first: when the oracle credit is known, which estimator is closer to it? That answer should inform later closed-loop experiments rather than be hidden inside them.

## 7 Results

The canonical matrix uses 20 seeds and 18 fixed cases across horizon length, group size, critic budget, observability, sparse-reward regimes, and a designed counterexample. Critic-style TD has the higher mean correlation in 17 of 18 cases. A confidence-interval reading is more cautious: 16 cases are clearly critic-favorable, 1 is a near tie, and 1 is clearly group-favorable. The mean critic-minus-group correlation is 0.420. Table 2 reports representative cases; the full CSV and JSON are committed in `results/`.

**Interpretation.** The evidence supports the PPO-favorable hypothesis in this controlled toy: when progress state is observable and critic coverage is adequate, critic-style advantages track token-level oracle credit much better than response-level group scalars. The result also protects against dogma. In the coverage-limited case, the confidence interval crosses zero; in the blind and undercovered case, group-relative terminal outcome information is better than a poor value table.

### Critic minus group correlation by case

Positive values favor critic-style TD; negative values favor group-relative advantage.

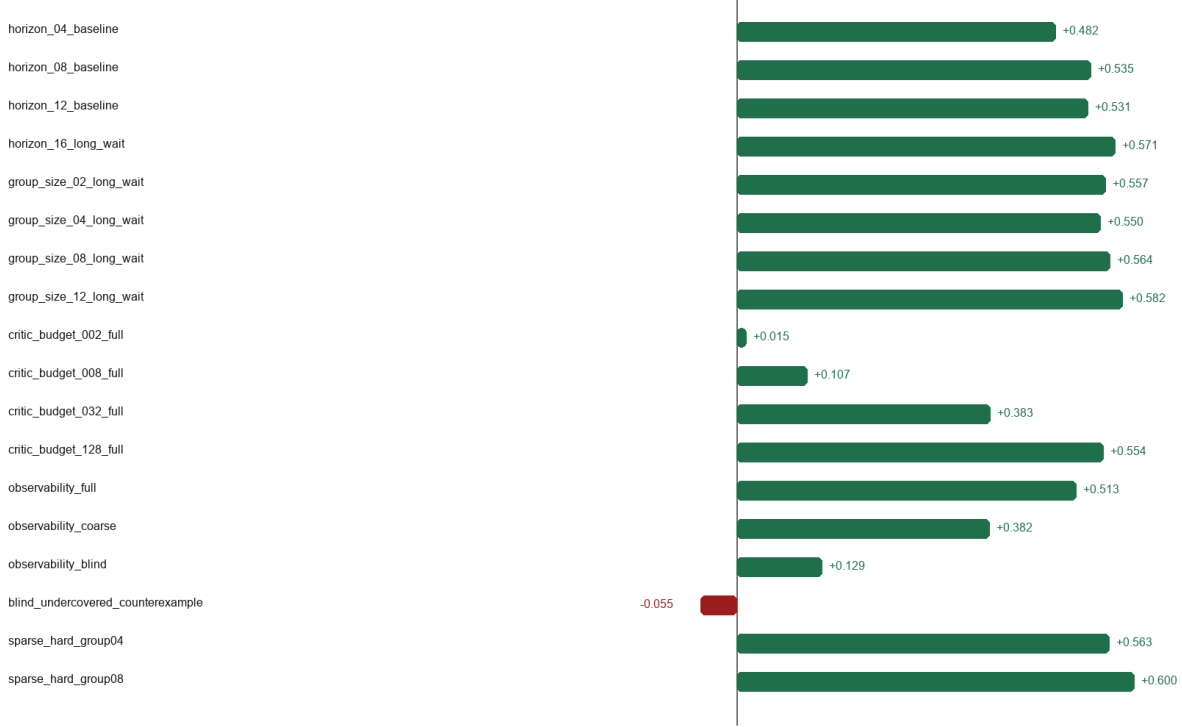


Figure 1: Mean critic-minus-group oracle-advantage correlation by case. Positive values favor critic-style temporal-difference advantages.

Table 2: Selected rows from the 20-seed deep matrix. Delta is critic  $r$  minus group  $r$ ; CI is a 95% across-seed interval for delta.

| Case                              | Group $r$ | Critic $r$ | Delta  | CI          | Reading      |
|-----------------------------------|-----------|------------|--------|-------------|--------------|
| horizon_04_baseline               | 0.495     | 0.977      | 0.482  | $\pm 0.011$ | critic clear |
| horizon_16_long_wait              | 0.293     | 0.865      | 0.571  | $\pm 0.013$ | critic clear |
| critic_budget_002_full            | 0.352     | 0.367      | 0.015  | $\pm 0.027$ | near tie     |
| critic_budget_128_full            | 0.352     | 0.906      | 0.554  | $\pm 0.017$ | critic clear |
| observability_blind               | 0.353     | 0.482      | 0.129  | $\pm 0.017$ | critic clear |
| blind_undercovered_counterexample | 0.510     | 0.455      | -0.055 | $\pm 0.015$ | group clear  |
| sparse_hard_group08               | 0.310     | 0.910      | 0.600  | $\pm 0.008$ | critic clear |



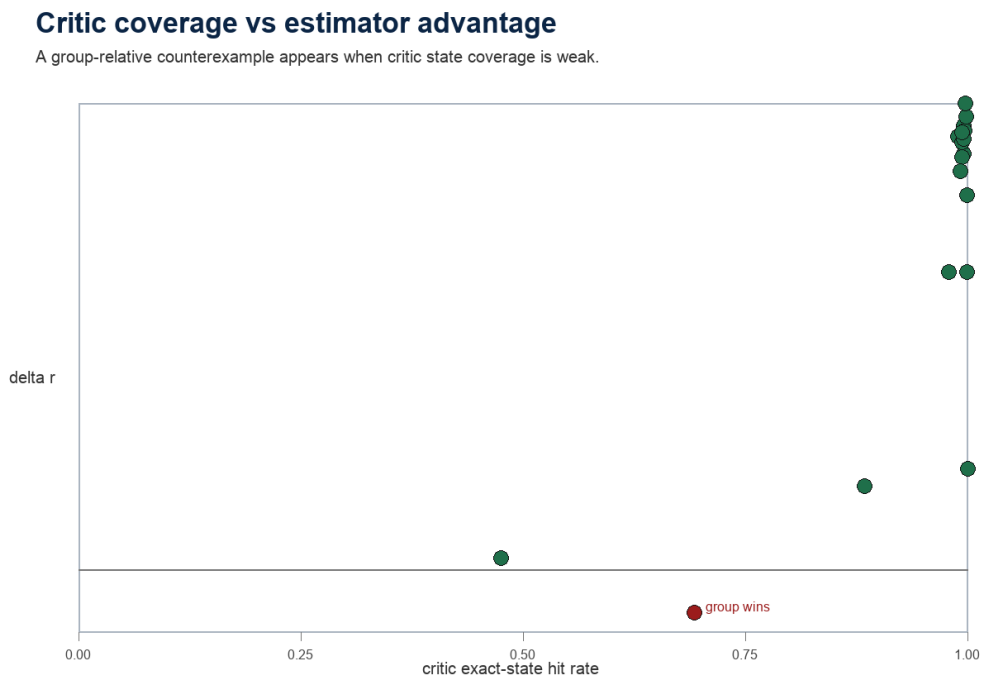


Figure 2: Critic state coverage versus estimator advantage. The group-relative counterexample occurs when the critic is blind and undercovered.

## 8 Mechanism Analysis

**Horizon length.** The horizon cases show the central intuition. As trajectories lengthen and wait tokens accumulate, a response-level scalar is asked to explain increasingly heterogeneous token roles. The group-relative estimator can still rank complete rollouts, but that does not mean it assigns good token credit. The critic-style estimator retains access to progress state and remaining horizon, so it can separate helpful actions from wait actions even when the final reward is delayed.

**Group size.** Increasing group size improves the quality of group statistics only when the additional samples create meaningful reward dispersion. In the long-wait cases, larger groups help the group-relative estimator somewhat, but the broadcast problem remains: every token in a completion inherits the same normalized outcome. In settings where verifier outcomes are sparse or clustered, larger groups can spend many tokens for little additional advantage information.

**Critic budget and coverage.** The critic-budget axis is the strongest warning against a simplistic “critics are better” claim. With only two training groups, the critic has an exact-state hit rate below one half and the confidence interval crosses zero. With more coverage, the critic delta grows sharply. In a neural setting, the analogous quantity is not table hit rate but value-model calibration under the current policy distribution. A poorly covered critic should be treated as a risk, not a badge of sophistication.

**Observability.** The observability cases show that a critic can still extract some value from coarse state, but a blind critic loses much of the temporal structure. The toy blind critic is intentionally simple; real long-horizon agents can become effectively blind for subtler reasons, including lossy compaction, missing tool state, stale memory, hidden environment variables, or value-model inputs that do not include the causal feature needed for future return.

**Counterexample.** The counterexample is not an embarrassment; it is the scientific control. If a paper about PPO versus GRPO cannot state when GRPO should win, it is not a method paper. In the blind-undercovered case, terminal group outcomes carry more usable information than a weak value table. The resulting group win is small but clear under the confidence-interval rule.

## 9 Which Method Is Better?

The answer is conditional, and the condition is about information. PPO-style critic methods are better candidates when future return is predictable from prefix state, when rollouts contain mixed token roles, and when the critic has enough coverage to be calibrated. GRPO-style methods are better candidates when rewards are cheap, verifiable, and meaningfully dispersed within prompt groups, or when the critic would be blind, stale, or too expensive. OPD and OPSD are better candidates when the available dense signal is a teacher or privileged self-teacher rather than an external reward.

This suggests a hybrid research program. Use rejection sampling and GRPO-like RL to find successes when verifiers are reliable. Use value or process models where temporal credit is the bottleneck. Use OPD/OPSD to distill verified or teacher-improved behaviors into cheaper policies. Report all of these with generated-token, verifier, teacher, critic, and wall-clock costs. The strongest method is unlikely to be a single objective used everywhere; it is more likely a pipeline whose components are selected by the information available at each training stage.

## 10 Failure Modes and Testable Predictions

**PPO/critic failures.** Critics can overfit reward artifacts, become stale after policy drift, collapse on long sparse trajectories, or assign confident but wrong temporal credit.

**GRPO failures.** GRPO can suffer zero-variance groups, low-dispersion amplification, response length bias, scalar reward broadcast over harmful tokens, and group composition confounding prompt difficulty with policy quality [4, 6, 8, 13, 16].

**OPD/OPSD failures.** Teacher/student thinking patterns may be incompatible; the teacher may offer no new information; on-policy rollouts may inflate in length and truncate; and privileged self-teacher traces may leak solution structure [3, 5, 15].

**Predictions.** Increasing horizon length and no-op padding should reduce the alignment between response-level group advantages and process labels. Increasing group size should help GRPO only when it increases meaningful reward dispersion. Adding process rewards should narrow the PPO/GRPO credit gap. Degrading critic observability should produce regimes where GRPO wins. OPD/OPSD should be most token-efficient when teacher signals are novel and rollout lengths are controlled.

## 11 Threats to Validity

**Toy dynamics.** The toy has a small finite state, a known oracle, and a tabular critic. This is deliberately unlike frontier post-training. The benefit is interpretability; the cost is that neural generalization, distribution shift, optimizer dynamics, and reward-model error are not measured.

**Estimator versus optimizer.** The experiment compares advantage estimators, not complete PPO and GRPO training loops. A full optimizer could amplify or dampen estimator differences through clipping, KL schedules, entropy, minibatching, and policy parameterization. This paper therefore supports a mechanism hypothesis, not a leaderboard conclusion.

**Simplified GRPO baseline.** The group-relative baseline is intentionally simple. Future comparisons should include leave-one-out baselines, length-normalized objectives, Dr. GRPO-style corrections, DAPO-style dynamic sampling, token-level process rewards, and low-dispersion safeguards. A stronger GRPO family member may narrow the gap in some regimes.

**Simplified critic.** The critic is tabular and trained from returns-to-go. Neural value models can generalize across states, but they can also hallucinate value structure. The toy’s exact-state hit rate is a proxy for coverage, not a complete model of critic reliability.

**External validity.** The GLM-5.2 discussion is based on a public vendor report and local text snapshot, not independent reproduction of its training stack. It should be read as evidence that the design pressure exists in industry systems, not as proof that critic-based PPO caused a benchmark gain.

## 12 Reproducibility

The core experiment code has no external Python dependencies. The canonical checks are:

```
python3 -m unittest discover -s tests
./scripts/run_smoke.sh
python3 -m experiments.deep_matrix \
  --output-json results/deep_matrix_20seed.json \
  --output-csv results/deep_matrix_20seed.csv \
  --output-md results/deep_matrix_20seed.md \
  --figures-dir results/figures
./scripts/build_latex_paper.sh
```

The LaTeX paper is built with Tectonic when available and emitted as the full paper PDF artifact under `public/`.

## 13 Limitations

The toy environment has a finite known state and a tabular critic. Real post-training adds neural approximation, KL controllers, reward-model errors, tool failures, distributed rollout systems, stale policies, optimizer details, and data contamination risks. The GRPO baseline is intentionally simple; the next benchmark should include leave-one-out baselines, length-normalized variants, DAPO/Dr. GRPO-style corrections, process rewards, and dynamic sampling before making stronger claims.

## 14 Roadmap for Limes Labs

**nanoGPT scale.** The next controlled step is a tiny neural policy where the true environment is still known but the critic must generalize. This can test whether tabular coverage results survive function approximation.

**AutoResearch and agentic tasks.** For agentic research workflows, the key artifact is a trace schema that records tool state, compaction boundaries, blocked actions, reward calls, and process labels. Without that schema, value modeling and group-relative normalization cannot be audited fairly.

**EuroBench and Parameter Golf.** Benchmark integrations should report cost-normalized improvement, not only final score. Parameter Golf is especially useful for comparing memory multipliers: a critic may improve credit assignment but violate a deployment budget. EuroBench-style tasks can test whether long-horizon behavior transfers from toy state to real language and tool use.

## 15 Conclusion

For long heterogeneous trajectories, critic-based PPO is a strong candidate when value targets are learnable and adequately covered. GRPO remains attractive when group reward contrast is reliable and critic modeling is too costly or poorly observed. OPD and OPSD are complementary dense-supervision tools that can transfer behaviors on student-visited states, but they should not

be conflated with reward optimization. The right public research program is therefore not “PPO always beats GRPO,” but a set of falsifiable boundaries for when temporal credit is worth its cost.

## A Formal Objective Notes

**Broadcasted group advantages.** In a response-level group-relative update, a completion-level reward statistic is often applied to every trainable token in that completion. Let  $g_i = (R_i - \mu_G)/(\sigma_G + \epsilon)$  be the group-normalized scalar for completion  $i$ . The token update is then proportional to  $g_i \nabla_{\theta} \log \pi_{\theta}(a_{i,t} | s_{i,t})$  for many  $t$ . This is efficient, but it implicitly assumes that all trainable tokens in the completion should move in the same reward direction. Long-horizon traces violate that assumption whenever success depends on a small subset of steps and the rest are neutral or harmful.

**Temporal-difference advantages.** A critic-style estimator replaces one response-level scalar with a local difference between predicted future return before and after an action. In the toy, this is a tabular one-step TD estimate. In language-model agents, the same idea may be implemented with value heads, separate value models, process reward models, or learned subgoal predictors. The central requirement is that the state representation contains enough information to make future return predictable.

**Distillation losses.** The OPD/OPSD losses in the main text are written as KL objectives because the teacher signal is a distribution over next tokens. Other implementations may use sequence-level filtering, hidden-state matching, preference-filtered teacher traces, or mixed supervised/RL objectives. These variants can be useful without changing the taxonomy: they are dense imitation or self-imitation signals rather than direct policy-gradient reward optimization.

## B Toy Environment Details

**State.** The state contains prompt identity, hidden progress, remaining horizon, and the features visible to the critic variant. The full critic observes enough to distinguish progress states. The coarse critic sees compressed state. The blind critic intentionally loses the feature needed for reliable future-return prediction.

**Actions.** The action set is deliberately small. **help** increases progress, **harm** decreases it, and **wait** consumes budget. This small action space makes exact dynamic programming feasible and makes the meaning of credit assignment visible. A language-model trace has richer tokens, but the causal categories are recognizable: helpful substeps, harmful detours, and neutral filler.

**Reward.** The terminal verifier is sparse and thresholded. This makes the group-relative estimator plausible, because rollouts can be ranked by final success, but also creates the credit problem: the reward is delayed until after many actions.

**Oracle advantage.** The oracle advantage uses exact values under known dynamics. This is the scientific anchor of the paper. Without it, a method comparison could only say which estimator produced higher downstream score under one optimizer. With it, we can ask whether an estimator points in the same direction as the true token-level credit.

## C Case Design

The 18 fixed cases are grouped by axis. Horizon cases increase trace length and wait-token exposure. Group-size cases vary the number of completions used for within-prompt normalization. Critic-budget cases vary how much data the value table sees before evaluation. Observability cases degrade the state available to the critic. Sparse-reward cases make terminal success harder. The counterexample combines blindness and low coverage.

Table 3: Axis-level summary generated from the canonical matrix.

| Axis           | Cases | Group $r$ | Critic $r$ | Delta  | Clear critic | Near tie | Clear group |
|----------------|-------|-----------|------------|--------|--------------|----------|-------------|
| counterexample | 1     | 0.510     | 0.455      | -0.055 | 0            | 0        | 1           |
| critic_budget  | 4     | 0.352     | 0.617      | 0.265  | 3            | 1        | 0           |
| group_size     | 4     | 0.305     | 0.868      | 0.563  | 4            | 0        | 0           |
| horizon        | 4     | 0.379     | 0.908      | 0.530  | 4            | 0        | 0           |
| observability  | 3     | 0.353     | 0.695      | 0.342  | 3            | 0        | 0           |
| sparse_reward  | 2     | 0.296     | 0.878      | 0.582  | 2            | 0        | 0           |

## D Full Case Summary

Table 4 gives the full 18-case summary. It is generated from the canonical matrix JSON rather than hand copied.

Table 4: Full 20-seed case summary. Delta is critic  $r$  minus group  $r$ .

| Case                 | Axis           | CI read  | Group $r$ | Critic $r$ | Delta  | CI          | Hit   | Wait  |
|----------------------|----------------|----------|-----------|------------|--------|-------------|-------|-------|
| horizon 04 baseline  | horizon        | critic   | 0.495     | 0.977      | 0.482  | $\pm 0.011$ | 0.999 | 0.303 |
| horizon 08 baseline  | horizon        | critic   | 0.390     | 0.925      | 0.535  | $\pm 0.015$ | 0.996 | 0.365 |
| horizon 12 baseline  | horizon        | critic   | 0.335     | 0.866      | 0.531  | $\pm 0.012$ | 0.994 | 0.411 |
| horizon 16 long wait | horizon        | critic   | 0.293     | 0.865      | 0.571  | $\pm 0.013$ | 0.996 | 0.539 |
| group 02 long wait   | group          | critic   | 0.213     | 0.771      | 0.557  | $\pm 0.022$ | 0.989 | 0.505 |
| group 04 long wait   | group          | critic   | 0.301     | 0.851      | 0.550  | $\pm 0.014$ | 0.994 | 0.503 |
| group 08 long wait   | group          | critic   | 0.346     | 0.910      | 0.564  | $\pm 0.011$ | 0.997 | 0.503 |
| group 12 long wait   | group          | critic   | 0.358     | 0.940      | 0.582  | $\pm 0.007$ | 0.998 | 0.501 |
| budget 002 full      | budget         | near tie | 0.352     | 0.367      | 0.015  | $\pm 0.027$ | 0.476 | 0.388 |
| budget 008 full      | budget         | critic   | 0.352     | 0.459      | 0.107  | $\pm 0.027$ | 0.884 | 0.388 |
| budget 032 full      | budget         | critic   | 0.352     | 0.735      | 0.383  | $\pm 0.021$ | 0.979 | 0.388 |
| budget 128 full      | budget         | critic   | 0.352     | 0.906      | 0.554  | $\pm 0.017$ | 0.996 | 0.388 |
| obs full             | obs            | critic   | 0.352     | 0.864      | 0.513  | $\pm 0.020$ | 0.992 | 0.388 |
| obs coarse           | obs            | critic   | 0.356     | 0.738      | 0.382  | $\pm 0.015$ | 0.999 | 0.450 |
| obs blind            | obs            | critic   | 0.353     | 0.482      | 0.129  | $\pm 0.017$ | 1.000 | 0.444 |
| blind undercovered   | counterexample | group    | 0.510     | 0.455      | -0.055 | $\pm 0.015$ | 0.693 | 0.329 |
| sparse group04       | sparse         | critic   | 0.283     | 0.846      | 0.563  | $\pm 0.014$ | 0.994 | 0.417 |
| sparse group08       | sparse         | critic   | 0.310     | 0.910      | 0.600  | $\pm 0.008$ | 0.997 | 0.420 |

## E Raw Seed-Level Rows

Table 5 records every case-by-seed row used to compute the paper’s aggregate claims. Publishing these rows makes the mean-win and confidence-interval interpretation auditable. The table is long by design: there are 18 cases and 20 seeds, for 360 rows.

Table 5: Raw case-by-seed estimator-fidelity rows from the deep matrix.

| Case                 | Seed | Group $r$ | Critic $r$ | Delta | Hit   | Success | Wait  |
|----------------------|------|-----------|------------|-------|-------|---------|-------|
| horizon 04 baseline  | 11   | 0.514     | 0.974      | 0.460 | 0.998 | 0.422   | 0.290 |
| horizon 04 baseline  | 29   | 0.480     | 0.976      | 0.495 | 0.998 | 0.391   | 0.312 |
| horizon 04 baseline  | 47   | 0.477     | 0.982      | 0.505 | 1.000 | 0.396   | 0.286 |
| horizon 04 baseline  | 83   | 0.522     | 0.975      | 0.452 | 1.000 | 0.458   | 0.331 |
| horizon 04 baseline  | 131  | 0.471     | 0.971      | 0.500 | 1.000 | 0.391   | 0.297 |
| horizon 04 baseline  | 173  | 0.505     | 0.971      | 0.466 | 1.000 | 0.422   | 0.296 |
| horizon 04 baseline  | 211  | 0.480     | 0.986      | 0.506 | 1.000 | 0.422   | 0.310 |
| horizon 04 baseline  | 257  | 0.503     | 0.980      | 0.477 | 1.000 | 0.427   | 0.305 |
| horizon 04 baseline  | 307  | 0.493     | 0.985      | 0.492 | 1.000 | 0.438   | 0.317 |
| horizon 04 baseline  | 359  | 0.456     | 0.981      | 0.525 | 1.000 | 0.422   | 0.295 |
| horizon 04 baseline  | 401  | 0.526     | 0.981      | 0.456 | 1.000 | 0.438   | 0.298 |
| horizon 04 baseline  | 463  | 0.491     | 0.981      | 0.490 | 1.000 | 0.417   | 0.309 |
| horizon 04 baseline  | 509  | 0.494     | 0.983      | 0.489 | 1.000 | 0.385   | 0.294 |
| horizon 04 baseline  | 571  | 0.463     | 0.969      | 0.506 | 0.997 | 0.396   | 0.330 |
| horizon 04 baseline  | 631  | 0.471     | 0.977      | 0.506 | 0.997 | 0.474   | 0.265 |
| horizon 04 baseline  | 701  | 0.499     | 0.973      | 0.475 | 1.000 | 0.448   | 0.304 |
| horizon 04 baseline  | 761  | 0.492     | 0.945      | 0.453 | 0.998 | 0.443   | 0.299 |
| horizon 04 baseline  | 823  | 0.548     | 0.974      | 0.427 | 1.000 | 0.464   | 0.312 |
| horizon 04 baseline  | 887  | 0.510     | 0.985      | 0.475 | 1.000 | 0.422   | 0.301 |
| horizon 04 baseline  | 947  | 0.508     | 0.985      | 0.477 | 1.000 | 0.448   | 0.304 |
| horizon 08 baseline  | 11   | 0.410     | 0.898      | 0.488 | 0.995 | 0.542   | 0.361 |
| horizon 08 baseline  | 29   | 0.370     | 0.917      | 0.548 | 0.998 | 0.531   | 0.382 |
| horizon 08 baseline  | 47   | 0.391     | 0.900      | 0.509 | 0.996 | 0.542   | 0.395 |
| horizon 08 baseline  | 83   | 0.365     | 0.923      | 0.558 | 0.996 | 0.490   | 0.369 |
| horizon 08 baseline  | 131  | 0.404     | 0.890      | 0.486 | 0.992 | 0.552   | 0.362 |
| horizon 08 baseline  | 173  | 0.395     | 0.924      | 0.529 | 0.999 | 0.542   | 0.373 |
| horizon 08 baseline  | 211  | 0.402     | 0.919      | 0.517 | 0.994 | 0.547   | 0.347 |
| horizon 08 baseline  | 257  | 0.401     | 0.928      | 0.527 | 0.997 | 0.552   | 0.373 |
| horizon 08 baseline  | 307  | 0.386     | 0.945      | 0.559 | 1.000 | 0.583   | 0.364 |
| horizon 08 baseline  | 359  | 0.339     | 0.957      | 0.618 | 1.000 | 0.526   | 0.376 |
| horizon 08 baseline  | 401  | 0.410     | 0.900      | 0.491 | 0.986 | 0.516   | 0.334 |
| horizon 08 baseline  | 463  | 0.374     | 0.925      | 0.551 | 0.998 | 0.589   | 0.380 |
| horizon 08 baseline  | 509  | 0.390     | 0.919      | 0.529 | 0.998 | 0.599   | 0.369 |
| horizon 08 baseline  | 571  | 0.376     | 0.973      | 0.596 | 0.999 | 0.568   | 0.347 |
| horizon 08 baseline  | 631  | 0.379     | 0.947      | 0.568 | 1.000 | 0.510   | 0.341 |
| horizon 08 baseline  | 701  | 0.400     | 0.925      | 0.526 | 0.996 | 0.583   | 0.404 |
| horizon 08 baseline  | 761  | 0.380     | 0.916      | 0.536 | 0.996 | 0.547   | 0.350 |
| horizon 08 baseline  | 823  | 0.420     | 0.913      | 0.494 | 0.990 | 0.578   | 0.357 |
| horizon 08 baseline  | 887  | 0.408     | 0.937      | 0.530 | 0.995 | 0.604   | 0.364 |
| horizon 08 baseline  | 947  | 0.405     | 0.945      | 0.540 | 1.000 | 0.500   | 0.358 |
| horizon 12 baseline  | 11   | 0.314     | 0.868      | 0.554 | 0.999 | 0.609   | 0.415 |
| horizon 12 baseline  | 29   | 0.348     | 0.919      | 0.571 | 0.995 | 0.589   | 0.400 |
| horizon 12 baseline  | 47   | 0.321     | 0.870      | 0.549 | 0.997 | 0.677   | 0.417 |
| horizon 12 baseline  | 83   | 0.346     | 0.869      | 0.524 | 0.994 | 0.573   | 0.396 |
| horizon 12 baseline  | 131  | 0.339     | 0.878      | 0.540 | 0.994 | 0.604   | 0.397 |
| horizon 12 baseline  | 173  | 0.357     | 0.836      | 0.479 | 0.994 | 0.630   | 0.416 |
| horizon 12 baseline  | 211  | 0.337     | 0.860      | 0.523 | 0.996 | 0.667   | 0.390 |
| horizon 12 baseline  | 257  | 0.346     | 0.893      | 0.547 | 0.997 | 0.656   | 0.437 |
| horizon 12 baseline  | 307  | 0.324     | 0.877      | 0.554 | 0.992 | 0.667   | 0.412 |
| horizon 12 baseline  | 359  | 0.349     | 0.846      | 0.497 | 0.995 | 0.599   | 0.397 |
| horizon 12 baseline  | 401  | 0.349     | 0.870      | 0.521 | 0.993 | 0.630   | 0.412 |
| horizon 12 baseline  | 463  | 0.348     | 0.842      | 0.494 | 0.997 | 0.656   | 0.416 |
| horizon 12 baseline  | 509  | 0.330     | 0.887      | 0.557 | 0.992 | 0.661   | 0.425 |
| horizon 12 baseline  | 571  | 0.326     | 0.840      | 0.514 | 0.986 | 0.661   | 0.424 |
| horizon 12 baseline  | 631  | 0.342     | 0.847      | 0.505 | 0.990 | 0.604   | 0.392 |
| horizon 12 baseline  | 701  | 0.316     | 0.895      | 0.578 | 0.996 | 0.651   | 0.445 |
| horizon 12 baseline  | 761  | 0.349     | 0.884      | 0.536 | 0.990 | 0.625   | 0.405 |
| horizon 12 baseline  | 823  | 0.341     | 0.847      | 0.507 | 0.990 | 0.573   | 0.399 |
| horizon 12 baseline  | 887  | 0.314     | 0.852      | 0.538 | 0.998 | 0.646   | 0.402 |
| horizon 12 baseline  | 947  | 0.314     | 0.844      | 0.530 | 0.993 | 0.620   | 0.421 |
| horizon 16 long wait | 11   | 0.302     | 0.862      | 0.560 | 0.998 | 0.667   | 0.513 |
| horizon 16 long wait | 29   | 0.303     | 0.884      | 0.581 | 0.998 | 0.667   | 0.536 |
| horizon 16 long wait | 47   | 0.266     | 0.866      | 0.601 | 0.998 | 0.718   | 0.538 |
| horizon 16 long wait | 83   | 0.319     | 0.885      | 0.565 | 0.996 | 0.620   | 0.529 |



| Case                 | Seed | Group $r$ | Critic $r$ | Delta | Hit   | Success | Wait  |
|----------------------|------|-----------|------------|-------|-------|---------|-------|
| horizon 16 long wait | 131  | 0.292     | 0.871      | 0.579 | 0.995 | 0.671   | 0.542 |
| horizon 16 long wait | 173  | 0.297     | 0.869      | 0.572 | 0.997 | 0.606   | 0.552 |
| horizon 16 long wait | 211  | 0.292     | 0.875      | 0.583 | 0.996 | 0.718   | 0.546 |
| horizon 16 long wait | 257  | 0.299     | 0.865      | 0.566 | 0.991 | 0.694   | 0.532 |
| horizon 16 long wait | 307  | 0.264     | 0.909      | 0.645 | 0.998 | 0.671   | 0.539 |
| horizon 16 long wait | 359  | 0.287     | 0.847      | 0.560 | 0.994 | 0.741   | 0.546 |
| horizon 16 long wait | 401  | 0.291     | 0.809      | 0.517 | 0.993 | 0.718   | 0.539 |
| horizon 16 long wait | 463  | 0.299     | 0.860      | 0.561 | 0.997 | 0.662   | 0.552 |
| horizon 16 long wait | 509  | 0.306     | 0.875      | 0.568 | 0.995 | 0.662   | 0.545 |
| horizon 16 long wait | 571  | 0.297     | 0.826      | 0.529 | 0.994 | 0.657   | 0.547 |
| horizon 16 long wait | 631  | 0.305     | 0.840      | 0.534 | 0.995 | 0.667   | 0.517 |
| horizon 16 long wait | 701  | 0.291     | 0.845      | 0.554 | 0.996 | 0.713   | 0.527 |
| horizon 16 long wait | 761  | 0.316     | 0.878      | 0.562 | 0.997 | 0.736   | 0.562 |
| horizon 16 long wait | 823  | 0.287     | 0.866      | 0.579 | 0.995 | 0.657   | 0.532 |
| horizon 16 long wait | 887  | 0.272     | 0.895      | 0.623 | 0.998 | 0.713   | 0.539 |
| horizon 16 long wait | 947  | 0.279     | 0.868      | 0.589 | 0.998 | 0.694   | 0.545 |
| group 02 long wait   | 11   | 0.174     | 0.665      | 0.492 | 0.968 | 0.641   | 0.507 |
| group 02 long wait   | 29   | 0.227     | 0.774      | 0.547 | 0.991 | 0.719   | 0.496 |
| group 02 long wait   | 47   | 0.223     | 0.792      | 0.569 | 0.994 | 0.656   | 0.495 |
| group 02 long wait   | 83   | 0.245     | 0.709      | 0.464 | 0.987 | 0.719   | 0.487 |
| group 02 long wait   | 131  | 0.189     | 0.758      | 0.569 | 0.998 | 0.625   | 0.514 |
| group 02 long wait   | 173  | 0.212     | 0.794      | 0.582 | 0.995 | 0.734   | 0.499 |
| group 02 long wait   | 211  | 0.216     | 0.773      | 0.557 | 0.987 | 0.578   | 0.489 |
| group 02 long wait   | 257  | 0.236     | 0.775      | 0.539 | 0.986 | 0.641   | 0.515 |
| group 02 long wait   | 307  | 0.187     | 0.811      | 0.624 | 0.995 | 0.672   | 0.533 |
| group 02 long wait   | 359  | 0.212     | 0.699      | 0.487 | 0.980 | 0.625   | 0.453 |
| group 02 long wait   | 401  | 0.195     | 0.720      | 0.525 | 0.996 | 0.672   | 0.528 |
| group 02 long wait   | 463  | 0.184     | 0.805      | 0.621 | 0.993 | 0.609   | 0.496 |
| group 02 long wait   | 509  | 0.223     | 0.754      | 0.532 | 0.979 | 0.656   | 0.515 |
| group 02 long wait   | 571  | 0.247     | 0.878      | 0.631 | 0.986 | 0.656   | 0.536 |
| group 02 long wait   | 631  | 0.214     | 0.777      | 0.563 | 0.998 | 0.641   | 0.492 |
| group 02 long wait   | 701  | 0.263     | 0.748      | 0.485 | 0.978 | 0.609   | 0.488 |
| group 02 long wait   | 761  | 0.225     | 0.783      | 0.558 | 0.998 | 0.672   | 0.490 |
| group 02 long wait   | 823  | 0.207     | 0.795      | 0.588 | 0.994 | 0.688   | 0.498 |
| group 02 long wait   | 887  | 0.203     | 0.774      | 0.571 | 0.987 | 0.734   | 0.537 |
| group 02 long wait   | 947  | 0.187     | 0.827      | 0.641 | 0.991 | 0.703   | 0.542 |
| group 04 long wait   | 11   | 0.273     | 0.788      | 0.515 | 0.991 | 0.680   | 0.488 |
| group 04 long wait   | 29   | 0.305     | 0.861      | 0.557 | 0.998 | 0.734   | 0.495 |
| group 04 long wait   | 47   | 0.265     | 0.794      | 0.529 | 0.988 | 0.656   | 0.481 |
| group 04 long wait   | 83   | 0.300     | 0.874      | 0.574 | 1.000 | 0.664   | 0.503 |
| group 04 long wait   | 131  | 0.323     | 0.867      | 0.544 | 0.998 | 0.625   | 0.524 |
| group 04 long wait   | 173  | 0.333     | 0.796      | 0.464 | 0.985 | 0.656   | 0.465 |
| group 04 long wait   | 211  | 0.291     | 0.846      | 0.555 | 0.999 | 0.609   | 0.507 |
| group 04 long wait   | 257  | 0.308     | 0.875      | 0.567 | 0.996 | 0.664   | 0.533 |
| group 04 long wait   | 307  | 0.331     | 0.879      | 0.548 | 0.991 | 0.641   | 0.495 |
| group 04 long wait   | 359  | 0.289     | 0.859      | 0.570 | 0.993 | 0.609   | 0.471 |
| group 04 long wait   | 401  | 0.298     | 0.909      | 0.611 | 0.999 | 0.656   | 0.544 |
| group 04 long wait   | 463  | 0.293     | 0.880      | 0.587 | 0.995 | 0.570   | 0.487 |
| group 04 long wait   | 509  | 0.300     | 0.825      | 0.525 | 0.992 | 0.617   | 0.516 |
| group 04 long wait   | 571  | 0.321     | 0.879      | 0.558 | 0.988 | 0.688   | 0.514 |
| group 04 long wait   | 631  | 0.303     | 0.841      | 0.538 | 0.998 | 0.680   | 0.485 |
| group 04 long wait   | 701  | 0.291     | 0.876      | 0.585 | 0.997 | 0.688   | 0.517 |
| group 04 long wait   | 761  | 0.296     | 0.871      | 0.575 | 1.000 | 0.688   | 0.491 |
| group 04 long wait   | 823  | 0.296     | 0.857      | 0.561 | 0.982 | 0.664   | 0.508 |
| group 04 long wait   | 887  | 0.282     | 0.799      | 0.517 | 0.995 | 0.672   | 0.508 |
| group 04 long wait   | 947  | 0.330     | 0.847      | 0.517 | 0.997 | 0.633   | 0.516 |
| group 08 long wait   | 11   | 0.337     | 0.903      | 0.566 | 0.997 | 0.656   | 0.506 |
| group 08 long wait   | 29   | 0.345     | 0.915      | 0.569 | 0.998 | 0.652   | 0.493 |
| group 08 long wait   | 47   | 0.301     | 0.907      | 0.607 | 0.998 | 0.680   | 0.506 |
| group 08 long wait   | 83   | 0.356     | 0.931      | 0.575 | 0.997 | 0.625   | 0.460 |
| group 08 long wait   | 131  | 0.373     | 0.931      | 0.558 | 0.996 | 0.637   | 0.522 |
| group 08 long wait   | 173  | 0.353     | 0.865      | 0.512 | 0.996 | 0.633   | 0.497 |
| group 08 long wait   | 211  | 0.357     | 0.927      | 0.570 | 0.998 | 0.621   | 0.503 |
| group 08 long wait   | 257  | 0.344     | 0.938      | 0.595 | 0.998 | 0.617   | 0.505 |
| group 08 long wait   | 307  | 0.333     | 0.925      | 0.592 | 0.998 | 0.664   | 0.511 |
| group 08 long wait   | 359  | 0.350     | 0.891      | 0.541 | 0.995 | 0.660   | 0.505 |

| Case               | Seed | Group $r$ | Critic $r$ | Delta  | Hit   | Success | Wait  |
|--------------------|------|-----------|------------|--------|-------|---------|-------|
| group 08 long wait | 401  | 0.343     | 0.924      | 0.581  | 0.998 | 0.660   | 0.515 |
| group 08 long wait | 463  | 0.340     | 0.900      | 0.560  | 0.999 | 0.609   | 0.518 |
| group 08 long wait | 509  | 0.333     | 0.925      | 0.592  | 0.996 | 0.691   | 0.503 |
| group 08 long wait | 571  | 0.333     | 0.867      | 0.534  | 0.994 | 0.672   | 0.526 |
| group 08 long wait | 631  | 0.366     | 0.904      | 0.538  | 0.997 | 0.609   | 0.517 |
| group 08 long wait | 701  | 0.342     | 0.899      | 0.556  | 0.998 | 0.668   | 0.505 |
| group 08 long wait | 761  | 0.343     | 0.945      | 0.602  | 0.998 | 0.586   | 0.494 |
| group 08 long wait | 823  | 0.346     | 0.887      | 0.541  | 0.989 | 0.617   | 0.484 |
| group 08 long wait | 887  | 0.369     | 0.914      | 0.545  | 0.997 | 0.613   | 0.484 |
| group 08 long wait | 947  | 0.350     | 0.900      | 0.550  | 0.996 | 0.660   | 0.513 |
| group 12 long wait | 11   | 0.365     | 0.924      | 0.559  | 0.999 | 0.646   | 0.496 |
| group 12 long wait | 29   | 0.367     | 0.935      | 0.568  | 0.999 | 0.633   | 0.491 |
| group 12 long wait | 47   | 0.338     | 0.958      | 0.620  | 0.999 | 0.659   | 0.511 |
| group 12 long wait | 83   | 0.372     | 0.939      | 0.568  | 0.999 | 0.674   | 0.498 |
| group 12 long wait | 131  | 0.353     | 0.958      | 0.605  | 1.000 | 0.638   | 0.531 |
| group 12 long wait | 173  | 0.355     | 0.948      | 0.593  | 0.998 | 0.690   | 0.490 |
| group 12 long wait | 211  | 0.351     | 0.948      | 0.596  | 0.998 | 0.617   | 0.488 |
| group 12 long wait | 257  | 0.352     | 0.940      | 0.588  | 0.999 | 0.607   | 0.511 |
| group 12 long wait | 307  | 0.367     | 0.943      | 0.576  | 1.000 | 0.654   | 0.502 |
| group 12 long wait | 359  | 0.348     | 0.938      | 0.590  | 0.998 | 0.667   | 0.493 |
| group 12 long wait | 401  | 0.363     | 0.937      | 0.574  | 0.997 | 0.674   | 0.502 |
| group 12 long wait | 463  | 0.356     | 0.919      | 0.564  | 0.997 | 0.646   | 0.520 |
| group 12 long wait | 509  | 0.352     | 0.942      | 0.590  | 0.996 | 0.635   | 0.490 |
| group 12 long wait | 571  | 0.368     | 0.928      | 0.560  | 0.996 | 0.628   | 0.511 |
| group 12 long wait | 631  | 0.371     | 0.937      | 0.566  | 0.998 | 0.656   | 0.507 |
| group 12 long wait | 701  | 0.367     | 0.945      | 0.578  | 1.000 | 0.693   | 0.513 |
| group 12 long wait | 761  | 0.344     | 0.934      | 0.590  | 0.998 | 0.659   | 0.500 |
| group 12 long wait | 823  | 0.348     | 0.947      | 0.599  | 0.997 | 0.617   | 0.486 |
| group 12 long wait | 887  | 0.356     | 0.942      | 0.585  | 0.998 | 0.635   | 0.496 |
| group 12 long wait | 947  | 0.363     | 0.942      | 0.579  | 0.999 | 0.656   | 0.494 |
| budget 002 full    | 11   | 0.358     | 0.415      | 0.057  | 0.467 | 0.573   | 0.392 |
| budget 002 full    | 29   | 0.395     | 0.405      | 0.009  | 0.476 | 0.620   | 0.400 |
| budget 002 full    | 47   | 0.342     | 0.330      | -0.012 | 0.536 | 0.609   | 0.385 |
| budget 002 full    | 83   | 0.308     | 0.251      | -0.056 | 0.439 | 0.630   | 0.409 |
| budget 002 full    | 131  | 0.319     | 0.340      | 0.021  | 0.369 | 0.646   | 0.381 |
| budget 002 full    | 173  | 0.328     | 0.441      | 0.113  | 0.514 | 0.562   | 0.381 |
| budget 002 full    | 211  | 0.363     | 0.299      | -0.064 | 0.501 | 0.630   | 0.398 |
| budget 002 full    | 257  | 0.359     | 0.451      | 0.091  | 0.459 | 0.516   | 0.375 |
| budget 002 full    | 307  | 0.357     | 0.267      | -0.090 | 0.522 | 0.594   | 0.380 |
| budget 002 full    | 359  | 0.313     | 0.414      | 0.101  | 0.376 | 0.568   | 0.364 |
| budget 002 full    | 401  | 0.385     | 0.304      | -0.081 | 0.506 | 0.573   | 0.374 |
| budget 002 full    | 463  | 0.381     | 0.484      | 0.102  | 0.454 | 0.557   | 0.407 |
| budget 002 full    | 509  | 0.329     | 0.358      | 0.029  | 0.489 | 0.604   | 0.381 |
| budget 002 full    | 571  | 0.381     | 0.379      | -0.003 | 0.432 | 0.599   | 0.397 |
| budget 002 full    | 631  | 0.364     | 0.306      | -0.058 | 0.470 | 0.562   | 0.388 |
| budget 002 full    | 701  | 0.318     | 0.368      | 0.050  | 0.543 | 0.594   | 0.406 |
| budget 002 full    | 761  | 0.329     | 0.375      | 0.046  | 0.503 | 0.651   | 0.413 |
| budget 002 full    | 823  | 0.387     | 0.409      | 0.022  | 0.389 | 0.615   | 0.392 |
| budget 002 full    | 887  | 0.377     | 0.368      | -0.009 | 0.528 | 0.609   | 0.360 |
| budget 002 full    | 947  | 0.342     | 0.383      | 0.040  | 0.539 | 0.599   | 0.372 |
| budget 008 full    | 11   | 0.358     | 0.471      | 0.114  | 0.868 | 0.573   | 0.392 |
| budget 008 full    | 29   | 0.395     | 0.536      | 0.141  | 0.873 | 0.620   | 0.400 |
| budget 008 full    | 47   | 0.342     | 0.507      | 0.165  | 0.913 | 0.609   | 0.385 |
| budget 008 full    | 83   | 0.308     | 0.470      | 0.163  | 0.856 | 0.630   | 0.409 |
| budget 008 full    | 131  | 0.319     | 0.395      | 0.076  | 0.878 | 0.646   | 0.381 |
| budget 008 full    | 173  | 0.328     | 0.567      | 0.239  | 0.885 | 0.562   | 0.381 |
| budget 008 full    | 211  | 0.363     | 0.466      | 0.104  | 0.846 | 0.630   | 0.398 |
| budget 008 full    | 257  | 0.359     | 0.517      | 0.158  | 0.917 | 0.516   | 0.375 |
| budget 008 full    | 307  | 0.357     | 0.399      | 0.042  | 0.926 | 0.594   | 0.380 |
| budget 008 full    | 359  | 0.313     | 0.530      | 0.217  | 0.872 | 0.568   | 0.364 |
| budget 008 full    | 401  | 0.385     | 0.480      | 0.096  | 0.885 | 0.573   | 0.374 |
| budget 008 full    | 463  | 0.381     | 0.455      | 0.074  | 0.958 | 0.557   | 0.407 |
| budget 008 full    | 509  | 0.329     | 0.435      | 0.106  | 0.913 | 0.604   | 0.381 |
| budget 008 full    | 571  | 0.381     | 0.465      | 0.084  | 0.875 | 0.599   | 0.397 |
| budget 008 full    | 631  | 0.364     | 0.353      | -0.012 | 0.828 | 0.562   | 0.388 |
| budget 008 full    | 701  | 0.318     | 0.422      | 0.104  | 0.884 | 0.594   | 0.406 |

| Case            | Seed | Group $r$ | Critic $r$ | Delta | Hit   | Success | Wait  |
|-----------------|------|-----------|------------|-------|-------|---------|-------|
| budget 008 full | 761  | 0.329     | 0.348      | 0.020 | 0.847 | 0.651   | 0.413 |
| budget 008 full | 823  | 0.387     | 0.429      | 0.042 | 0.849 | 0.615   | 0.392 |
| budget 008 full | 887  | 0.377     | 0.501      | 0.124 | 0.903 | 0.609   | 0.360 |
| budget 008 full | 947  | 0.342     | 0.434      | 0.092 | 0.902 | 0.599   | 0.372 |
| budget 032 full | 11   | 0.358     | 0.715      | 0.357 | 0.975 | 0.573   | 0.392 |
| budget 032 full | 29   | 0.395     | 0.761      | 0.365 | 0.974 | 0.620   | 0.400 |
| budget 032 full | 47   | 0.342     | 0.749      | 0.407 | 0.987 | 0.609   | 0.385 |
| budget 032 full | 83   | 0.308     | 0.709      | 0.402 | 0.975 | 0.630   | 0.409 |
| budget 032 full | 131  | 0.319     | 0.715      | 0.396 | 0.974 | 0.646   | 0.381 |
| budget 032 full | 173  | 0.328     | 0.828      | 0.500 | 0.980 | 0.562   | 0.381 |
| budget 032 full | 211  | 0.363     | 0.667      | 0.305 | 0.971 | 0.630   | 0.398 |
| budget 032 full | 257  | 0.359     | 0.795      | 0.436 | 0.993 | 0.516   | 0.375 |
| budget 032 full | 307  | 0.357     | 0.708      | 0.351 | 0.983 | 0.594   | 0.380 |
| budget 032 full | 359  | 0.313     | 0.780      | 0.467 | 0.982 | 0.568   | 0.364 |
| budget 032 full | 401  | 0.385     | 0.691      | 0.306 | 0.973 | 0.573   | 0.374 |
| budget 032 full | 463  | 0.381     | 0.757      | 0.375 | 0.987 | 0.557   | 0.407 |
| budget 032 full | 509  | 0.329     | 0.672      | 0.343 | 0.981 | 0.604   | 0.381 |
| budget 032 full | 571  | 0.381     | 0.717      | 0.336 | 0.979 | 0.599   | 0.397 |
| budget 032 full | 631  | 0.364     | 0.720      | 0.356 | 0.961 | 0.562   | 0.388 |
| budget 032 full | 701  | 0.318     | 0.700      | 0.382 | 0.969 | 0.594   | 0.406 |
| budget 032 full | 761  | 0.329     | 0.700      | 0.372 | 0.973 | 0.651   | 0.413 |
| budget 032 full | 823  | 0.387     | 0.798      | 0.411 | 0.993 | 0.615   | 0.392 |
| budget 032 full | 887  | 0.377     | 0.771      | 0.394 | 0.985 | 0.609   | 0.360 |
| budget 032 full | 947  | 0.342     | 0.740      | 0.397 | 0.984 | 0.599   | 0.372 |
| budget 128 full | 11   | 0.358     | 0.907      | 0.549 | 0.994 | 0.573   | 0.392 |
| budget 128 full | 29   | 0.395     | 0.884      | 0.489 | 0.996 | 0.620   | 0.400 |
| budget 128 full | 47   | 0.342     | 0.909      | 0.567 | 0.997 | 0.609   | 0.385 |
| budget 128 full | 83   | 0.308     | 0.896      | 0.588 | 0.996 | 0.630   | 0.409 |
| budget 128 full | 131  | 0.319     | 0.899      | 0.580 | 0.992 | 0.646   | 0.381 |
| budget 128 full | 173  | 0.328     | 0.942      | 0.614 | 1.000 | 0.562   | 0.381 |
| budget 128 full | 211  | 0.363     | 0.873      | 0.510 | 0.993 | 0.630   | 0.398 |
| budget 128 full | 257  | 0.359     | 0.917      | 0.557 | 0.998 | 0.516   | 0.375 |
| budget 128 full | 307  | 0.357     | 0.880      | 0.523 | 0.993 | 0.594   | 0.380 |
| budget 128 full | 359  | 0.313     | 0.938      | 0.625 | 0.997 | 0.568   | 0.364 |
| budget 128 full | 401  | 0.385     | 0.905      | 0.521 | 0.992 | 0.573   | 0.374 |
| budget 128 full | 463  | 0.381     | 0.915      | 0.533 | 0.996 | 0.557   | 0.407 |
| budget 128 full | 509  | 0.329     | 0.923      | 0.594 | 0.993 | 0.604   | 0.381 |
| budget 128 full | 571  | 0.381     | 0.880      | 0.498 | 0.997 | 0.599   | 0.397 |
| budget 128 full | 631  | 0.364     | 0.881      | 0.516 | 0.987 | 0.562   | 0.388 |
| budget 128 full | 701  | 0.318     | 0.909      | 0.591 | 0.999 | 0.594   | 0.406 |
| budget 128 full | 761  | 0.329     | 0.897      | 0.568 | 0.997 | 0.651   | 0.413 |
| budget 128 full | 823  | 0.387     | 0.935      | 0.549 | 0.996 | 0.615   | 0.392 |
| budget 128 full | 887  | 0.377     | 0.914      | 0.537 | 0.998 | 0.609   | 0.360 |
| budget 128 full | 947  | 0.342     | 0.919      | 0.577 | 0.998 | 0.599   | 0.372 |
| obs full        | 11   | 0.358     | 0.843      | 0.485 | 0.991 | 0.573   | 0.392 |
| obs full        | 29   | 0.395     | 0.856      | 0.460 | 0.992 | 0.620   | 0.400 |
| obs full        | 47   | 0.342     | 0.867      | 0.525 | 0.996 | 0.609   | 0.385 |
| obs full        | 83   | 0.308     | 0.872      | 0.564 | 0.993 | 0.630   | 0.409 |
| obs full        | 131  | 0.319     | 0.843      | 0.524 | 0.984 | 0.646   | 0.381 |
| obs full        | 173  | 0.328     | 0.935      | 0.607 | 1.000 | 0.562   | 0.381 |
| obs full        | 211  | 0.363     | 0.835      | 0.473 | 0.987 | 0.630   | 0.398 |
| obs full        | 257  | 0.359     | 0.871      | 0.512 | 0.995 | 0.516   | 0.375 |
| obs full        | 307  | 0.357     | 0.868      | 0.510 | 0.991 | 0.594   | 0.380 |
| obs full        | 359  | 0.313     | 0.890      | 0.577 | 0.993 | 0.568   | 0.364 |
| obs full        | 401  | 0.385     | 0.809      | 0.424 | 0.989 | 0.573   | 0.374 |
| obs full        | 463  | 0.381     | 0.870      | 0.489 | 0.994 | 0.557   | 0.407 |
| obs full        | 509  | 0.329     | 0.893      | 0.564 | 0.989 | 0.604   | 0.381 |
| obs full        | 571  | 0.381     | 0.836      | 0.454 | 0.993 | 0.599   | 0.397 |
| obs full        | 631  | 0.364     | 0.845      | 0.481 | 0.984 | 0.562   | 0.388 |
| obs full        | 701  | 0.318     | 0.831      | 0.513 | 0.989 | 0.594   | 0.406 |
| obs full        | 761  | 0.329     | 0.859      | 0.531 | 0.996 | 0.651   | 0.413 |
| obs full        | 823  | 0.387     | 0.891      | 0.504 | 0.996 | 0.615   | 0.392 |
| obs full        | 887  | 0.377     | 0.894      | 0.517 | 0.995 | 0.609   | 0.360 |
| obs full        | 947  | 0.342     | 0.881      | 0.539 | 0.996 | 0.599   | 0.372 |
| obs coarse      | 11   | 0.361     | 0.701      | 0.341 | 1.000 | 0.568   | 0.440 |
| obs coarse      | 29   | 0.378     | 0.710      | 0.331 | 1.000 | 0.630   | 0.453 |

| Case               | Seed | Group $r$ | Critic $r$ | Delta  | Hit   | Success | Wait  |
|--------------------|------|-----------|------------|--------|-------|---------|-------|
| obs coarse         | 47   | 0.351     | 0.702      | 0.352  | 1.000 | 0.646   | 0.453 |
| obs coarse         | 83   | 0.321     | 0.773      | 0.452  | 1.000 | 0.578   | 0.456 |
| obs coarse         | 131  | 0.344     | 0.727      | 0.382  | 1.000 | 0.615   | 0.434 |
| obs coarse         | 173  | 0.355     | 0.728      | 0.373  | 1.000 | 0.609   | 0.446 |
| obs coarse         | 211  | 0.366     | 0.735      | 0.370  | 0.997 | 0.630   | 0.453 |
| obs coarse         | 257  | 0.380     | 0.767      | 0.386  | 1.000 | 0.547   | 0.441 |
| obs coarse         | 307  | 0.342     | 0.710      | 0.368  | 1.000 | 0.620   | 0.452 |
| obs coarse         | 359  | 0.328     | 0.760      | 0.432  | 1.000 | 0.599   | 0.461 |
| obs coarse         | 401  | 0.371     | 0.758      | 0.387  | 1.000 | 0.625   | 0.443 |
| obs coarse         | 463  | 0.367     | 0.703      | 0.336  | 1.000 | 0.604   | 0.457 |
| obs coarse         | 509  | 0.308     | 0.750      | 0.442  | 1.000 | 0.635   | 0.441 |
| obs coarse         | 571  | 0.379     | 0.760      | 0.382  | 1.000 | 0.625   | 0.464 |
| obs coarse         | 631  | 0.357     | 0.757      | 0.400  | 0.998 | 0.557   | 0.433 |
| obs coarse         | 701  | 0.345     | 0.714      | 0.369  | 0.998 | 0.656   | 0.477 |
| obs coarse         | 761  | 0.359     | 0.707      | 0.348  | 0.997 | 0.620   | 0.464 |
| obs coarse         | 823  | 0.375     | 0.761      | 0.386  | 1.000 | 0.573   | 0.445 |
| obs coarse         | 887  | 0.380     | 0.756      | 0.375  | 0.999 | 0.594   | 0.425 |
| obs coarse         | 947  | 0.347     | 0.784      | 0.437  | 1.000 | 0.609   | 0.454 |
| obs blind          | 11   | 0.358     | 0.400      | 0.042  | 1.000 | 0.568   | 0.442 |
| obs blind          | 29   | 0.376     | 0.501      | 0.125  | 1.000 | 0.630   | 0.444 |
| obs blind          | 47   | 0.342     | 0.423      | 0.081  | 1.000 | 0.651   | 0.454 |
| obs blind          | 83   | 0.321     | 0.459      | 0.138  | 1.000 | 0.589   | 0.457 |
| obs blind          | 131  | 0.340     | 0.463      | 0.124  | 1.000 | 0.615   | 0.426 |
| obs blind          | 173  | 0.360     | 0.485      | 0.125  | 1.000 | 0.599   | 0.442 |
| obs blind          | 211  | 0.355     | 0.519      | 0.164  | 1.000 | 0.615   | 0.450 |
| obs blind          | 257  | 0.384     | 0.487      | 0.103  | 1.000 | 0.542   | 0.435 |
| obs blind          | 307  | 0.336     | 0.417      | 0.081  | 1.000 | 0.615   | 0.441 |
| obs blind          | 359  | 0.330     | 0.521      | 0.190  | 1.000 | 0.604   | 0.444 |
| obs blind          | 401  | 0.360     | 0.456      | 0.096  | 1.000 | 0.620   | 0.437 |
| obs blind          | 463  | 0.372     | 0.477      | 0.104  | 1.000 | 0.594   | 0.451 |
| obs blind          | 509  | 0.297     | 0.486      | 0.189  | 1.000 | 0.641   | 0.436 |
| obs blind          | 571  | 0.377     | 0.532      | 0.155  | 1.000 | 0.620   | 0.454 |
| obs blind          | 631  | 0.349     | 0.491      | 0.143  | 1.000 | 0.562   | 0.426 |
| obs blind          | 701  | 0.341     | 0.487      | 0.145  | 1.000 | 0.641   | 0.470 |
| obs blind          | 761  | 0.352     | 0.486      | 0.134  | 1.000 | 0.630   | 0.468 |
| obs blind          | 823  | 0.375     | 0.551      | 0.177  | 1.000 | 0.562   | 0.435 |
| obs blind          | 887  | 0.374     | 0.500      | 0.126  | 1.000 | 0.604   | 0.423 |
| obs blind          | 947  | 0.351     | 0.499      | 0.148  | 1.000 | 0.604   | 0.449 |
| blind undercovered | 11   | 0.518     | 0.460      | -0.058 | 0.694 | 0.398   | 0.313 |
| blind undercovered | 29   | 0.472     | 0.496      | 0.024  | 0.697 | 0.414   | 0.324 |
| blind undercovered | 47   | 0.518     | 0.450      | -0.068 | 0.683 | 0.434   | 0.330 |
| blind undercovered | 83   | 0.512     | 0.413      | -0.099 | 0.689 | 0.430   | 0.357 |
| blind undercovered | 131  | 0.525     | 0.464      | -0.061 | 0.681 | 0.434   | 0.315 |
| blind undercovered | 173  | 0.525     | 0.505      | -0.020 | 0.680 | 0.469   | 0.316 |
| blind undercovered | 211  | 0.508     | 0.494      | -0.015 | 0.670 | 0.461   | 0.327 |
| blind undercovered | 257  | 0.502     | 0.432      | -0.070 | 0.699 | 0.406   | 0.309 |
| blind undercovered | 307  | 0.536     | 0.478      | -0.057 | 0.697 | 0.477   | 0.339 |
| blind undercovered | 359  | 0.525     | 0.420      | -0.105 | 0.706 | 0.426   | 0.315 |
| blind undercovered | 401  | 0.502     | 0.408      | -0.094 | 0.704 | 0.398   | 0.334 |
| blind undercovered | 463  | 0.503     | 0.471      | -0.031 | 0.692 | 0.391   | 0.352 |
| blind undercovered | 509  | 0.525     | 0.495      | -0.030 | 0.705 | 0.434   | 0.348 |
| blind undercovered | 571  | 0.509     | 0.496      | -0.014 | 0.692 | 0.418   | 0.362 |
| blind undercovered | 631  | 0.473     | 0.432      | -0.040 | 0.683 | 0.418   | 0.297 |
| blind undercovered | 701  | 0.513     | 0.463      | -0.050 | 0.686 | 0.430   | 0.323 |
| blind undercovered | 761  | 0.498     | 0.401      | -0.098 | 0.698 | 0.438   | 0.316 |
| blind undercovered | 823  | 0.520     | 0.448      | -0.072 | 0.703 | 0.449   | 0.332 |
| blind undercovered | 887  | 0.496     | 0.447      | -0.049 | 0.706 | 0.426   | 0.343 |
| blind undercovered | 947  | 0.513     | 0.419      | -0.094 | 0.688 | 0.449   | 0.322 |
| sparse group04     | 11   | 0.282     | 0.832      | 0.551  | 0.994 | 0.486   | 0.407 |
| sparse group04     | 29   | 0.290     | 0.848      | 0.558  | 0.997 | 0.528   | 0.442 |
| sparse group04     | 47   | 0.301     | 0.829      | 0.528  | 0.991 | 0.472   | 0.413 |
| sparse group04     | 83   | 0.246     | 0.853      | 0.607  | 0.998 | 0.458   | 0.432 |
| sparse group04     | 131  | 0.320     | 0.838      | 0.518  | 0.996 | 0.479   | 0.433 |
| sparse group04     | 173  | 0.309     | 0.880      | 0.571  | 0.994 | 0.472   | 0.399 |
| sparse group04     | 211  | 0.286     | 0.852      | 0.566  | 0.993 | 0.493   | 0.404 |
| sparse group04     | 257  | 0.274     | 0.858      | 0.584  | 0.996 | 0.431   | 0.419 |

| Case           | Seed | Group $r$ | Critic $r$ | Delta | Hit   | Success | Wait  |
|----------------|------|-----------|------------|-------|-------|---------|-------|
| sparse group04 | 307  | 0.303     | 0.857      | 0.555 | 0.991 | 0.403   | 0.412 |
| sparse group04 | 359  | 0.244     | 0.820      | 0.575 | 0.992 | 0.493   | 0.412 |
| sparse group04 | 401  | 0.278     | 0.817      | 0.539 | 0.988 | 0.479   | 0.417 |
| sparse group04 | 463  | 0.305     | 0.870      | 0.564 | 0.996 | 0.507   | 0.435 |
| sparse group04 | 509  | 0.296     | 0.864      | 0.568 | 0.986 | 0.424   | 0.439 |
| sparse group04 | 571  | 0.251     | 0.836      | 0.586 | 0.996 | 0.472   | 0.408 |
| sparse group04 | 631  | 0.287     | 0.845      | 0.558 | 0.993 | 0.451   | 0.399 |
| sparse group04 | 701  | 0.262     | 0.835      | 0.573 | 0.998 | 0.472   | 0.435 |
| sparse group04 | 761  | 0.286     | 0.781      | 0.495 | 0.987 | 0.493   | 0.429 |
| sparse group04 | 823  | 0.259     | 0.908      | 0.649 | 0.996 | 0.500   | 0.410 |
| sparse group04 | 887  | 0.287     | 0.861      | 0.574 | 0.994 | 0.479   | 0.412 |
| sparse group04 | 947  | 0.289     | 0.829      | 0.540 | 0.998 | 0.500   | 0.385 |
| sparse group08 | 11   | 0.337     | 0.924      | 0.587 | 0.999 | 0.497   | 0.409 |
| sparse group08 | 29   | 0.296     | 0.897      | 0.601 | 1.000 | 0.455   | 0.434 |
| sparse group08 | 47   | 0.324     | 0.894      | 0.570 | 0.995 | 0.462   | 0.425 |
| sparse group08 | 83   | 0.302     | 0.915      | 0.613 | 0.996 | 0.455   | 0.417 |
| sparse group08 | 131  | 0.307     | 0.931      | 0.624 | 0.998 | 0.458   | 0.430 |
| sparse group08 | 173  | 0.312     | 0.911      | 0.599 | 0.995 | 0.469   | 0.415 |
| sparse group08 | 211  | 0.299     | 0.921      | 0.622 | 0.999 | 0.455   | 0.405 |
| sparse group08 | 257  | 0.318     | 0.901      | 0.582 | 0.999 | 0.462   | 0.414 |
| sparse group08 | 307  | 0.302     | 0.912      | 0.610 | 0.998 | 0.417   | 0.419 |
| sparse group08 | 359  | 0.324     | 0.894      | 0.570 | 0.995 | 0.472   | 0.419 |
| sparse group08 | 401  | 0.296     | 0.885      | 0.589 | 0.996 | 0.479   | 0.417 |
| sparse group08 | 463  | 0.302     | 0.931      | 0.629 | 0.999 | 0.507   | 0.429 |
| sparse group08 | 509  | 0.318     | 0.923      | 0.605 | 0.994 | 0.476   | 0.428 |
| sparse group08 | 571  | 0.302     | 0.916      | 0.614 | 0.999 | 0.427   | 0.415 |
| sparse group08 | 631  | 0.321     | 0.904      | 0.583 | 0.997 | 0.465   | 0.417 |
| sparse group08 | 701  | 0.316     | 0.903      | 0.587 | 0.999 | 0.490   | 0.434 |
| sparse group08 | 761  | 0.298     | 0.914      | 0.616 | 0.997 | 0.479   | 0.434 |
| sparse group08 | 823  | 0.311     | 0.903      | 0.591 | 0.996 | 0.444   | 0.402 |
| sparse group08 | 887  | 0.306     | 0.916      | 0.610 | 0.997 | 0.476   | 0.419 |
| sparse group08 | 947  | 0.309     | 0.911      | 0.602 | 0.998 | 0.465   | 0.416 |

## F Raw Error and Dispersion Rows

Table 6 records the calibrated-MSE side of the same matrix. These rows help distinguish two different claims: a method can have higher correlation with oracle credit while also changing calibrated error, zero-variance group frequency, or success-rate mix.

Table 6: Raw case-by-seed calibrated-error rows from the deep matrix.

| Case                | Seed | Group MSE | Critic MSE | Delta MSE | Zero-var | Wait  | Success |
|---------------------|------|-----------|------------|-----------|----------|-------|---------|
| horizon 04 baseline | 11   | 0.050     | 0.004      | -0.046    | 0.000    | 0.290 | 0.422   |
| horizon 04 baseline | 29   | 0.053     | 0.003      | -0.050    | 0.000    | 0.312 | 0.391   |
| horizon 04 baseline | 47   | 0.047     | 0.002      | -0.045    | 0.000    | 0.286 | 0.396   |
| horizon 04 baseline | 83   | 0.044     | 0.003      | -0.041    | 0.000    | 0.331 | 0.458   |
| horizon 04 baseline | 131  | 0.047     | 0.003      | -0.044    | 0.000    | 0.297 | 0.391   |
| horizon 04 baseline | 173  | 0.049     | 0.004      | -0.046    | 0.000    | 0.296 | 0.422   |
| horizon 04 baseline | 211  | 0.053     | 0.002      | -0.051    | 0.000    | 0.310 | 0.422   |
| horizon 04 baseline | 257  | 0.054     | 0.003      | -0.051    | 0.000    | 0.305 | 0.427   |
| horizon 04 baseline | 307  | 0.044     | 0.002      | -0.042    | 0.000    | 0.317 | 0.438   |
| horizon 04 baseline | 359  | 0.058     | 0.003      | -0.055    | 0.000    | 0.295 | 0.422   |
| horizon 04 baseline | 401  | 0.051     | 0.003      | -0.048    | 0.000    | 0.298 | 0.438   |
| horizon 04 baseline | 463  | 0.051     | 0.003      | -0.048    | 0.000    | 0.309 | 0.417   |
| horizon 04 baseline | 509  | 0.054     | 0.002      | -0.052    | 0.000    | 0.294 | 0.385   |
| horizon 04 baseline | 571  | 0.052     | 0.004      | -0.048    | 0.000    | 0.330 | 0.396   |
| horizon 04 baseline | 631  | 0.056     | 0.003      | -0.053    | 0.000    | 0.265 | 0.474   |
| horizon 04 baseline | 701  | 0.052     | 0.004      | -0.049    | 0.000    | 0.304 | 0.448   |
| horizon 04 baseline | 761  | 0.047     | 0.007      | -0.040    | 0.000    | 0.299 | 0.443   |
| horizon 04 baseline | 823  | 0.045     | 0.003      | -0.042    | 0.000    | 0.312 | 0.464   |
| horizon 04 baseline | 887  | 0.049     | 0.002      | -0.047    | 0.000    | 0.301 | 0.422   |

| Case                 | Seed | Group MSE | Critic MSE | Delta MSE | Zero-var | Wait  | Success |
|----------------------|------|-----------|------------|-----------|----------|-------|---------|
| horizon 04 baseline  | 947  | 0.050     | 0.002      | -0.048    | 0.000    | 0.304 | 0.448   |
| horizon 08 baseline  | 11   | 0.034     | 0.008      | -0.026    | 0.000    | 0.361 | 0.542   |
| horizon 08 baseline  | 29   | 0.036     | 0.007      | -0.029    | 0.000    | 0.382 | 0.531   |
| horizon 08 baseline  | 47   | 0.036     | 0.008      | -0.028    | 0.000    | 0.395 | 0.542   |
| horizon 08 baseline  | 83   | 0.036     | 0.006      | -0.030    | 0.000    | 0.369 | 0.490   |
| horizon 08 baseline  | 131  | 0.030     | 0.007      | -0.022    | 0.000    | 0.362 | 0.552   |
| horizon 08 baseline  | 173  | 0.038     | 0.007      | -0.031    | 0.000    | 0.373 | 0.542   |
| horizon 08 baseline  | 211  | 0.038     | 0.007      | -0.031    | 0.000    | 0.347 | 0.547   |
| horizon 08 baseline  | 257  | 0.035     | 0.006      | -0.029    | 0.000    | 0.373 | 0.552   |
| horizon 08 baseline  | 307  | 0.034     | 0.004      | -0.030    | 0.000    | 0.364 | 0.583   |
| horizon 08 baseline  | 359  | 0.039     | 0.004      | -0.035    | 0.000    | 0.376 | 0.526   |
| horizon 08 baseline  | 401  | 0.035     | 0.008      | -0.027    | 0.000    | 0.334 | 0.516   |
| horizon 08 baseline  | 463  | 0.037     | 0.006      | -0.031    | 0.000    | 0.380 | 0.589   |
| horizon 08 baseline  | 509  | 0.034     | 0.006      | -0.028    | 0.000    | 0.369 | 0.599   |
| horizon 08 baseline  | 571  | 0.040     | 0.003      | -0.037    | 0.000    | 0.347 | 0.568   |
| horizon 08 baseline  | 631  | 0.035     | 0.004      | -0.031    | 0.000    | 0.341 | 0.510   |
| horizon 08 baseline  | 701  | 0.033     | 0.006      | -0.027    | 0.000    | 0.404 | 0.583   |
| horizon 08 baseline  | 761  | 0.037     | 0.007      | -0.030    | 0.000    | 0.350 | 0.547   |
| horizon 08 baseline  | 823  | 0.031     | 0.006      | -0.025    | 0.000    | 0.357 | 0.578   |
| horizon 08 baseline  | 887  | 0.032     | 0.005      | -0.028    | 0.000    | 0.364 | 0.604   |
| horizon 08 baseline  | 947  | 0.037     | 0.005      | -0.032    | 0.000    | 0.358 | 0.500   |
| horizon 12 baseline  | 11   | 0.024     | 0.007      | -0.017    | 0.000    | 0.415 | 0.609   |
| horizon 12 baseline  | 29   | 0.028     | 0.005      | -0.023    | 0.000    | 0.400 | 0.589   |
| horizon 12 baseline  | 47   | 0.022     | 0.006      | -0.016    | 0.000    | 0.417 | 0.677   |
| horizon 12 baseline  | 83   | 0.027     | 0.007      | -0.019    | 0.000    | 0.396 | 0.573   |
| horizon 12 baseline  | 131  | 0.026     | 0.007      | -0.019    | 0.000    | 0.397 | 0.604   |
| horizon 12 baseline  | 173  | 0.026     | 0.009      | -0.017    | 0.000    | 0.416 | 0.630   |
| horizon 12 baseline  | 211  | 0.025     | 0.007      | -0.018    | 0.000    | 0.390 | 0.667   |
| horizon 12 baseline  | 257  | 0.023     | 0.005      | -0.017    | 0.000    | 0.437 | 0.656   |
| horizon 12 baseline  | 307  | 0.025     | 0.006      | -0.018    | 0.000    | 0.412 | 0.667   |
| horizon 12 baseline  | 359  | 0.024     | 0.008      | -0.016    | 0.000    | 0.397 | 0.599   |
| horizon 12 baseline  | 401  | 0.025     | 0.007      | -0.018    | 0.000    | 0.412 | 0.630   |
| horizon 12 baseline  | 463  | 0.022     | 0.007      | -0.015    | 0.000    | 0.416 | 0.656   |
| horizon 12 baseline  | 509  | 0.025     | 0.006      | -0.019    | 0.000    | 0.425 | 0.661   |
| horizon 12 baseline  | 571  | 0.024     | 0.008      | -0.016    | 0.000    | 0.424 | 0.661   |
| horizon 12 baseline  | 631  | 0.028     | 0.009      | -0.019    | 0.000    | 0.392 | 0.604   |
| horizon 12 baseline  | 701  | 0.024     | 0.005      | -0.019    | 0.000    | 0.445 | 0.651   |
| horizon 12 baseline  | 761  | 0.025     | 0.006      | -0.019    | 0.000    | 0.405 | 0.625   |
| horizon 12 baseline  | 823  | 0.027     | 0.009      | -0.018    | 0.000    | 0.399 | 0.573   |
| horizon 12 baseline  | 887  | 0.025     | 0.008      | -0.017    | 0.000    | 0.402 | 0.646   |
| horizon 12 baseline  | 947  | 0.025     | 0.008      | -0.017    | 0.000    | 0.421 | 0.620   |
| horizon 16 long wait | 11   | 0.021     | 0.006      | -0.015    | 0.000    | 0.513 | 0.667   |
| horizon 16 long wait | 29   | 0.018     | 0.004      | -0.014    | 0.000    | 0.536 | 0.667   |
| horizon 16 long wait | 47   | 0.017     | 0.005      | -0.013    | 0.000    | 0.538 | 0.718   |
| horizon 16 long wait | 83   | 0.020     | 0.005      | -0.016    | 0.000    | 0.529 | 0.620   |
| horizon 16 long wait | 131  | 0.019     | 0.005      | -0.014    | 0.000    | 0.542 | 0.671   |
| horizon 16 long wait | 173  | 0.019     | 0.005      | -0.014    | 0.000    | 0.552 | 0.606   |
| horizon 16 long wait | 211  | 0.019     | 0.005      | -0.014    | 0.000    | 0.546 | 0.718   |
| horizon 16 long wait | 257  | 0.017     | 0.005      | -0.012    | 0.000    | 0.532 | 0.694   |
| horizon 16 long wait | 307  | 0.023     | 0.004      | -0.019    | 0.000    | 0.539 | 0.671   |
| horizon 16 long wait | 359  | 0.016     | 0.005      | -0.011    | 0.000    | 0.546 | 0.741   |
| horizon 16 long wait | 401  | 0.017     | 0.006      | -0.011    | 0.000    | 0.539 | 0.718   |
| horizon 16 long wait | 463  | 0.018     | 0.005      | -0.013    | 0.000    | 0.552 | 0.662   |
| horizon 16 long wait | 509  | 0.019     | 0.005      | -0.014    | 0.000    | 0.545 | 0.662   |
| horizon 16 long wait | 571  | 0.018     | 0.006      | -0.012    | 0.000    | 0.547 | 0.657   |
| horizon 16 long wait | 631  | 0.019     | 0.006      | -0.013    | 0.000    | 0.517 | 0.667   |
| horizon 16 long wait | 701  | 0.017     | 0.005      | -0.012    | 0.000    | 0.527 | 0.713   |
| horizon 16 long wait | 761  | 0.015     | 0.004      | -0.011    | 0.000    | 0.562 | 0.736   |
| horizon 16 long wait | 823  | 0.022     | 0.006      | -0.016    | 0.000    | 0.532 | 0.657   |
| horizon 16 long wait | 887  | 0.020     | 0.004      | -0.016    | 0.000    | 0.539 | 0.713   |
| horizon 16 long wait | 947  | 0.018     | 0.005      | -0.013    | 0.000    | 0.545 | 0.694   |
| group 02 long wait   | 11   | 0.024     | 0.014      | -0.010    | 0.031    | 0.507 | 0.641   |
| group 02 long wait   | 29   | 0.024     | 0.010      | -0.014    | 0.094    | 0.496 | 0.719   |
| group 02 long wait   | 47   | 0.025     | 0.010      | -0.015    | 0.062    | 0.495 | 0.656   |
| group 02 long wait   | 83   | 0.022     | 0.011      | -0.010    | 0.125    | 0.487 | 0.719   |
| group 02 long wait   | 131  | 0.024     | 0.010      | -0.013    | 0.094    | 0.514 | 0.625   |

| Case               | Seed | Group MSE | Critic MSE | Delta MSE | Zero-var | Wait  | Success |
|--------------------|------|-----------|------------|-----------|----------|-------|---------|
| group 02 long wait | 173  | 0.024     | 0.009      | -0.015    | 0.125    | 0.499 | 0.734   |
| group 02 long wait | 211  | 0.025     | 0.010      | -0.014    | 0.031    | 0.489 | 0.578   |
| group 02 long wait | 257  | 0.024     | 0.010      | -0.014    | 0.156    | 0.515 | 0.641   |
| group 02 long wait | 307  | 0.026     | 0.009      | -0.017    | 0.062    | 0.533 | 0.672   |
| group 02 long wait | 359  | 0.026     | 0.014      | -0.012    | 0.000    | 0.453 | 0.625   |
| group 02 long wait | 401  | 0.024     | 0.012      | -0.012    | 0.062    | 0.528 | 0.672   |
| group 02 long wait | 463  | 0.022     | 0.008      | -0.014    | 0.062    | 0.496 | 0.609   |
| group 02 long wait | 509  | 0.028     | 0.013      | -0.015    | 0.125    | 0.515 | 0.656   |
| group 02 long wait | 571  | 0.035     | 0.009      | -0.026    | 0.062    | 0.536 | 0.656   |
| group 02 long wait | 631  | 0.027     | 0.011      | -0.016    | 0.000    | 0.492 | 0.641   |
| group 02 long wait | 701  | 0.028     | 0.013      | -0.015    | 0.062    | 0.488 | 0.609   |
| group 02 long wait | 761  | 0.027     | 0.011      | -0.016    | 0.031    | 0.490 | 0.672   |
| group 02 long wait | 823  | 0.024     | 0.009      | -0.015    | 0.062    | 0.498 | 0.688   |
| group 02 long wait | 887  | 0.017     | 0.007      | -0.010    | 0.031    | 0.537 | 0.734   |
| group 02 long wait | 947  | 0.024     | 0.008      | -0.016    | 0.031    | 0.542 | 0.703   |
| group 04 long wait | 11   | 0.023     | 0.010      | -0.014    | 0.000    | 0.488 | 0.680   |
| group 04 long wait | 29   | 0.022     | 0.006      | -0.016    | 0.000    | 0.495 | 0.734   |
| group 04 long wait | 47   | 0.023     | 0.009      | -0.014    | 0.000    | 0.481 | 0.656   |
| group 04 long wait | 83   | 0.022     | 0.006      | -0.016    | 0.000    | 0.503 | 0.664   |
| group 04 long wait | 131  | 0.025     | 0.007      | -0.018    | 0.000    | 0.524 | 0.625   |
| group 04 long wait | 173  | 0.025     | 0.010      | -0.015    | 0.000    | 0.465 | 0.656   |
| group 04 long wait | 211  | 0.026     | 0.008      | -0.018    | 0.000    | 0.507 | 0.609   |
| group 04 long wait | 257  | 0.021     | 0.006      | -0.016    | 0.000    | 0.533 | 0.664   |
| group 04 long wait | 307  | 0.025     | 0.006      | -0.019    | 0.000    | 0.495 | 0.641   |
| group 04 long wait | 359  | 0.029     | 0.008      | -0.021    | 0.000    | 0.471 | 0.609   |
| group 04 long wait | 401  | 0.022     | 0.004      | -0.018    | 0.000    | 0.544 | 0.656   |
| group 04 long wait | 463  | 0.027     | 0.007      | -0.021    | 0.000    | 0.487 | 0.570   |
| group 04 long wait | 509  | 0.028     | 0.010      | -0.018    | 0.000    | 0.516 | 0.617   |
| group 04 long wait | 571  | 0.026     | 0.007      | -0.019    | 0.000    | 0.514 | 0.688   |
| group 04 long wait | 631  | 0.021     | 0.007      | -0.014    | 0.000    | 0.485 | 0.680   |
| group 04 long wait | 701  | 0.023     | 0.006      | -0.017    | 0.000    | 0.517 | 0.688   |
| group 04 long wait | 761  | 0.022     | 0.006      | -0.017    | 0.000    | 0.491 | 0.688   |
| group 04 long wait | 823  | 0.024     | 0.007      | -0.017    | 0.000    | 0.508 | 0.664   |
| group 04 long wait | 887  | 0.024     | 0.009      | -0.015    | 0.000    | 0.508 | 0.672   |
| group 04 long wait | 947  | 0.020     | 0.007      | -0.014    | 0.000    | 0.516 | 0.633   |
| group 08 long wait | 11   | 0.024     | 0.005      | -0.019    | 0.000    | 0.506 | 0.656   |
| group 08 long wait | 29   | 0.026     | 0.005      | -0.021    | 0.000    | 0.493 | 0.652   |
| group 08 long wait | 47   | 0.022     | 0.004      | -0.018    | 0.000    | 0.506 | 0.680   |
| group 08 long wait | 83   | 0.025     | 0.004      | -0.021    | 0.000    | 0.460 | 0.625   |
| group 08 long wait | 131  | 0.022     | 0.003      | -0.019    | 0.000    | 0.522 | 0.637   |
| group 08 long wait | 173  | 0.025     | 0.007      | -0.018    | 0.000    | 0.497 | 0.633   |
| group 08 long wait | 211  | 0.026     | 0.004      | -0.021    | 0.000    | 0.503 | 0.621   |
| group 08 long wait | 257  | 0.025     | 0.003      | -0.021    | 0.000    | 0.505 | 0.617   |
| group 08 long wait | 307  | 0.026     | 0.004      | -0.021    | 0.000    | 0.511 | 0.664   |
| group 08 long wait | 359  | 0.024     | 0.006      | -0.019    | 0.000    | 0.505 | 0.660   |
| group 08 long wait | 401  | 0.024     | 0.004      | -0.020    | 0.000    | 0.515 | 0.660   |
| group 08 long wait | 463  | 0.025     | 0.005      | -0.020    | 0.000    | 0.518 | 0.609   |
| group 08 long wait | 509  | 0.024     | 0.004      | -0.020    | 0.000    | 0.503 | 0.691   |
| group 08 long wait | 571  | 0.023     | 0.007      | -0.017    | 0.000    | 0.526 | 0.672   |
| group 08 long wait | 631  | 0.021     | 0.004      | -0.017    | 0.000    | 0.517 | 0.609   |
| group 08 long wait | 701  | 0.021     | 0.005      | -0.017    | 0.000    | 0.505 | 0.668   |
| group 08 long wait | 761  | 0.026     | 0.003      | -0.023    | 0.000    | 0.494 | 0.586   |
| group 08 long wait | 823  | 0.024     | 0.006      | -0.018    | 0.000    | 0.484 | 0.617   |
| group 08 long wait | 887  | 0.026     | 0.005      | -0.021    | 0.000    | 0.484 | 0.613   |
| group 08 long wait | 947  | 0.020     | 0.004      | -0.016    | 0.000    | 0.513 | 0.660   |
| group 12 long wait | 11   | 0.024     | 0.004      | -0.020    | 0.000    | 0.496 | 0.646   |
| group 12 long wait | 29   | 0.026     | 0.004      | -0.023    | 0.000    | 0.491 | 0.633   |
| group 12 long wait | 47   | 0.023     | 0.002      | -0.021    | 0.000    | 0.511 | 0.659   |
| group 12 long wait | 83   | 0.023     | 0.003      | -0.020    | 0.000    | 0.498 | 0.674   |
| group 12 long wait | 131  | 0.024     | 0.002      | -0.022    | 0.000    | 0.531 | 0.638   |
| group 12 long wait | 173  | 0.023     | 0.003      | -0.021    | 0.000    | 0.490 | 0.690   |
| group 12 long wait | 211  | 0.028     | 0.003      | -0.024    | 0.000    | 0.488 | 0.617   |
| group 12 long wait | 257  | 0.025     | 0.003      | -0.021    | 0.000    | 0.511 | 0.607   |
| group 12 long wait | 307  | 0.025     | 0.003      | -0.022    | 0.000    | 0.502 | 0.654   |
| group 12 long wait | 359  | 0.024     | 0.003      | -0.021    | 0.000    | 0.493 | 0.667   |
| group 12 long wait | 401  | 0.023     | 0.003      | -0.020    | 0.000    | 0.502 | 0.674   |

| Case               | Seed | Group MSE | Critic MSE | Delta MSE | Zero-var | Wait  | Success |
|--------------------|------|-----------|------------|-----------|----------|-------|---------|
| group 12 long wait | 463  | 0.024     | 0.004      | -0.020    | 0.000    | 0.520 | 0.646   |
| group 12 long wait | 509  | 0.025     | 0.003      | -0.022    | 0.000    | 0.490 | 0.635   |
| group 12 long wait | 571  | 0.024     | 0.004      | -0.020    | 0.000    | 0.511 | 0.628   |
| group 12 long wait | 631  | 0.021     | 0.003      | -0.018    | 0.000    | 0.507 | 0.656   |
| group 12 long wait | 701  | 0.021     | 0.003      | -0.018    | 0.000    | 0.513 | 0.693   |
| group 12 long wait | 761  | 0.024     | 0.003      | -0.020    | 0.000    | 0.500 | 0.659   |
| group 12 long wait | 823  | 0.026     | 0.003      | -0.023    | 0.000    | 0.486 | 0.617   |
| group 12 long wait | 887  | 0.026     | 0.003      | -0.023    | 0.000    | 0.496 | 0.635   |
| group 12 long wait | 947  | 0.023     | 0.003      | -0.020    | 0.000    | 0.494 | 0.656   |
| budget 002 full    | 11   | 0.029     | 0.027      | -0.001    | 0.000    | 0.392 | 0.573   |
| budget 002 full    | 29   | 0.028     | 0.028      | -0.000    | 0.000    | 0.400 | 0.620   |
| budget 002 full    | 47   | 0.029     | 0.029      | 0.000     | 0.000    | 0.385 | 0.609   |
| budget 002 full    | 83   | 0.027     | 0.028      | 0.001     | 0.000    | 0.409 | 0.630   |
| budget 002 full    | 131  | 0.028     | 0.028      | -0.000    | 0.000    | 0.381 | 0.646   |
| budget 002 full    | 173  | 0.036     | 0.033      | -0.004    | 0.000    | 0.381 | 0.562   |
| budget 002 full    | 211  | 0.028     | 0.029      | 0.001     | 0.000    | 0.398 | 0.630   |
| budget 002 full    | 257  | 0.034     | 0.031      | -0.003    | 0.000    | 0.375 | 0.516   |
| budget 002 full    | 307  | 0.029     | 0.031      | 0.002     | 0.000    | 0.380 | 0.594   |
| budget 002 full    | 359  | 0.035     | 0.032      | -0.003    | 0.000    | 0.364 | 0.568   |
| budget 002 full    | 401  | 0.030     | 0.032      | 0.002     | 0.000    | 0.374 | 0.573   |
| budget 002 full    | 463  | 0.033     | 0.029      | -0.003    | 0.000    | 0.407 | 0.557   |
| budget 002 full    | 509  | 0.030     | 0.030      | -0.001    | 0.000    | 0.381 | 0.604   |
| budget 002 full    | 571  | 0.031     | 0.031      | 0.000     | 0.000    | 0.397 | 0.599   |
| budget 002 full    | 631  | 0.031     | 0.032      | 0.001     | 0.000    | 0.388 | 0.562   |
| budget 002 full    | 701  | 0.032     | 0.031      | -0.001    | 0.000    | 0.406 | 0.594   |
| budget 002 full    | 761  | 0.026     | 0.025      | -0.001    | 0.000    | 0.413 | 0.651   |
| budget 002 full    | 823  | 0.027     | 0.026      | -0.001    | 0.000    | 0.392 | 0.615   |
| budget 002 full    | 887  | 0.031     | 0.032      | 0.000     | 0.000    | 0.360 | 0.609   |
| budget 002 full    | 947  | 0.033     | 0.032      | -0.001    | 0.000    | 0.372 | 0.599   |
| budget 008 full    | 11   | 0.029     | 0.026      | -0.003    | 0.000    | 0.392 | 0.573   |
| budget 008 full    | 29   | 0.028     | 0.024      | -0.004    | 0.000    | 0.400 | 0.620   |
| budget 008 full    | 47   | 0.029     | 0.024      | -0.005    | 0.000    | 0.385 | 0.609   |
| budget 008 full    | 83   | 0.027     | 0.024      | -0.004    | 0.000    | 0.409 | 0.630   |
| budget 008 full    | 131  | 0.028     | 0.026      | -0.002    | 0.000    | 0.381 | 0.646   |
| budget 008 full    | 173  | 0.036     | 0.027      | -0.009    | 0.000    | 0.381 | 0.562   |
| budget 008 full    | 211  | 0.028     | 0.025      | -0.003    | 0.000    | 0.398 | 0.630   |
| budget 008 full    | 257  | 0.034     | 0.028      | -0.005    | 0.000    | 0.375 | 0.516   |
| budget 008 full    | 307  | 0.029     | 0.028      | -0.001    | 0.000    | 0.380 | 0.594   |
| budget 008 full    | 359  | 0.035     | 0.028      | -0.007    | 0.000    | 0.364 | 0.568   |
| budget 008 full    | 401  | 0.030     | 0.027      | -0.003    | 0.000    | 0.374 | 0.573   |
| budget 008 full    | 463  | 0.033     | 0.030      | -0.002    | 0.000    | 0.407 | 0.557   |
| budget 008 full    | 509  | 0.030     | 0.028      | -0.003    | 0.000    | 0.381 | 0.604   |
| budget 008 full    | 571  | 0.031     | 0.028      | -0.003    | 0.000    | 0.397 | 0.599   |
| budget 008 full    | 631  | 0.031     | 0.031      | 0.000     | 0.000    | 0.388 | 0.562   |
| budget 008 full    | 701  | 0.032     | 0.029      | -0.003    | 0.000    | 0.406 | 0.594   |
| budget 008 full    | 761  | 0.026     | 0.026      | -0.000    | 0.000    | 0.413 | 0.651   |
| budget 008 full    | 823  | 0.027     | 0.026      | -0.001    | 0.000    | 0.392 | 0.615   |
| budget 008 full    | 887  | 0.031     | 0.027      | -0.004    | 0.000    | 0.360 | 0.609   |
| budget 008 full    | 947  | 0.033     | 0.030      | -0.003    | 0.000    | 0.372 | 0.599   |
| budget 032 full    | 11   | 0.029     | 0.016      | -0.013    | 0.000    | 0.392 | 0.573   |
| budget 032 full    | 29   | 0.028     | 0.014      | -0.014    | 0.000    | 0.400 | 0.620   |
| budget 032 full    | 47   | 0.029     | 0.014      | -0.014    | 0.000    | 0.385 | 0.609   |
| budget 032 full    | 83   | 0.027     | 0.015      | -0.012    | 0.000    | 0.409 | 0.630   |
| budget 032 full    | 131  | 0.028     | 0.015      | -0.013    | 0.000    | 0.381 | 0.646   |
| budget 032 full    | 173  | 0.036     | 0.013      | -0.023    | 0.000    | 0.381 | 0.562   |
| budget 032 full    | 211  | 0.028     | 0.018      | -0.010    | 0.000    | 0.398 | 0.630   |
| budget 032 full    | 257  | 0.034     | 0.014      | -0.019    | 0.000    | 0.375 | 0.516   |
| budget 032 full    | 307  | 0.029     | 0.017      | -0.013    | 0.000    | 0.380 | 0.594   |
| budget 032 full    | 359  | 0.035     | 0.015      | -0.020    | 0.000    | 0.364 | 0.568   |
| budget 032 full    | 401  | 0.030     | 0.018      | -0.012    | 0.000    | 0.374 | 0.573   |
| budget 032 full    | 463  | 0.033     | 0.016      | -0.016    | 0.000    | 0.407 | 0.557   |
| budget 032 full    | 509  | 0.030     | 0.019      | -0.012    | 0.000    | 0.381 | 0.604   |
| budget 032 full    | 571  | 0.031     | 0.018      | -0.013    | 0.000    | 0.397 | 0.599   |
| budget 032 full    | 631  | 0.031     | 0.017      | -0.014    | 0.000    | 0.388 | 0.562   |
| budget 032 full    | 701  | 0.032     | 0.018      | -0.014    | 0.000    | 0.406 | 0.594   |
| budget 032 full    | 761  | 0.026     | 0.015      | -0.011    | 0.000    | 0.413 | 0.651   |



| Case            | Seed | Group MSE | Critic MSE | Delta MSE | Zero-var | Wait  | Success |
|-----------------|------|-----------|------------|-----------|----------|-------|---------|
| budget 032 full | 823  | 0.027     | 0.011      | -0.015    | 0.000    | 0.392 | 0.615   |
| budget 032 full | 887  | 0.031     | 0.015      | -0.017    | 0.000    | 0.360 | 0.609   |
| budget 032 full | 947  | 0.033     | 0.017      | -0.016    | 0.000    | 0.372 | 0.599   |
| budget 128 full | 11   | 0.029     | 0.006      | -0.023    | 0.000    | 0.392 | 0.573   |
| budget 128 full | 29   | 0.028     | 0.007      | -0.021    | 0.000    | 0.400 | 0.620   |
| budget 128 full | 47   | 0.029     | 0.006      | -0.023    | 0.000    | 0.385 | 0.609   |
| budget 128 full | 83   | 0.027     | 0.006      | -0.021    | 0.000    | 0.409 | 0.630   |
| budget 128 full | 131  | 0.028     | 0.006      | -0.022    | 0.000    | 0.381 | 0.646   |
| budget 128 full | 173  | 0.036     | 0.005      | -0.032    | 0.000    | 0.381 | 0.562   |
| budget 128 full | 211  | 0.028     | 0.008      | -0.020    | 0.000    | 0.398 | 0.630   |
| budget 128 full | 257  | 0.034     | 0.006      | -0.027    | 0.000    | 0.375 | 0.516   |
| budget 128 full | 307  | 0.029     | 0.008      | -0.022    | 0.000    | 0.380 | 0.594   |
| budget 128 full | 359  | 0.035     | 0.005      | -0.030    | 0.000    | 0.364 | 0.568   |
| budget 128 full | 401  | 0.030     | 0.006      | -0.024    | 0.000    | 0.374 | 0.573   |
| budget 128 full | 463  | 0.033     | 0.006      | -0.027    | 0.000    | 0.407 | 0.557   |
| budget 128 full | 509  | 0.030     | 0.005      | -0.025    | 0.000    | 0.381 | 0.604   |
| budget 128 full | 571  | 0.031     | 0.008      | -0.023    | 0.000    | 0.397 | 0.599   |
| budget 128 full | 631  | 0.031     | 0.008      | -0.023    | 0.000    | 0.388 | 0.562   |
| budget 128 full | 701  | 0.032     | 0.006      | -0.026    | 0.000    | 0.406 | 0.594   |
| budget 128 full | 761  | 0.026     | 0.006      | -0.020    | 0.000    | 0.413 | 0.651   |
| budget 128 full | 823  | 0.027     | 0.004      | -0.023    | 0.000    | 0.392 | 0.615   |
| budget 128 full | 887  | 0.031     | 0.006      | -0.025    | 0.000    | 0.360 | 0.609   |
| budget 128 full | 947  | 0.033     | 0.006      | -0.027    | 0.000    | 0.372 | 0.599   |
| obs full        | 11   | 0.029     | 0.010      | -0.019    | 0.000    | 0.392 | 0.573   |
| obs full        | 29   | 0.028     | 0.009      | -0.019    | 0.000    | 0.400 | 0.620   |
| obs full        | 47   | 0.029     | 0.008      | -0.021    | 0.000    | 0.385 | 0.609   |
| obs full        | 83   | 0.027     | 0.007      | -0.020    | 0.000    | 0.409 | 0.630   |
| obs full        | 131  | 0.028     | 0.009      | -0.019    | 0.000    | 0.381 | 0.646   |
| obs full        | 173  | 0.036     | 0.005      | -0.031    | 0.000    | 0.381 | 0.562   |
| obs full        | 211  | 0.028     | 0.010      | -0.018    | 0.000    | 0.398 | 0.630   |
| obs full        | 257  | 0.034     | 0.009      | -0.024    | 0.000    | 0.375 | 0.516   |
| obs full        | 307  | 0.029     | 0.008      | -0.021    | 0.000    | 0.380 | 0.594   |
| obs full        | 359  | 0.035     | 0.008      | -0.027    | 0.000    | 0.364 | 0.568   |
| obs full        | 401  | 0.030     | 0.012      | -0.018    | 0.000    | 0.374 | 0.573   |
| obs full        | 463  | 0.033     | 0.009      | -0.024    | 0.000    | 0.407 | 0.557   |
| obs full        | 509  | 0.030     | 0.007      | -0.024    | 0.000    | 0.381 | 0.604   |
| obs full        | 571  | 0.031     | 0.011      | -0.020    | 0.000    | 0.397 | 0.599   |
| obs full        | 631  | 0.031     | 0.010      | -0.021    | 0.000    | 0.388 | 0.562   |
| obs full        | 701  | 0.032     | 0.011      | -0.021    | 0.000    | 0.406 | 0.594   |
| obs full        | 761  | 0.026     | 0.008      | -0.018    | 0.000    | 0.413 | 0.651   |
| obs full        | 823  | 0.027     | 0.007      | -0.020    | 0.000    | 0.392 | 0.615   |
| obs full        | 887  | 0.031     | 0.007      | -0.024    | 0.000    | 0.360 | 0.609   |
| obs full        | 947  | 0.033     | 0.008      | -0.024    | 0.000    | 0.372 | 0.599   |
| obs coarse      | 11   | 0.029     | 0.017      | -0.012    | 0.000    | 0.440 | 0.568   |
| obs coarse      | 29   | 0.029     | 0.017      | -0.012    | 0.000    | 0.453 | 0.630   |
| obs coarse      | 47   | 0.025     | 0.014      | -0.010    | 0.000    | 0.453 | 0.646   |
| obs coarse      | 83   | 0.031     | 0.014      | -0.017    | 0.000    | 0.456 | 0.578   |
| obs coarse      | 131  | 0.031     | 0.017      | -0.014    | 0.000    | 0.434 | 0.615   |
| obs coarse      | 173  | 0.031     | 0.016      | -0.014    | 0.000    | 0.446 | 0.609   |
| obs coarse      | 211  | 0.028     | 0.015      | -0.013    | 0.000    | 0.453 | 0.630   |
| obs coarse      | 257  | 0.033     | 0.016      | -0.017    | 0.000    | 0.441 | 0.547   |
| obs coarse      | 307  | 0.028     | 0.016      | -0.012    | 0.000    | 0.452 | 0.620   |
| obs coarse      | 359  | 0.031     | 0.015      | -0.016    | 0.000    | 0.461 | 0.599   |
| obs coarse      | 401  | 0.029     | 0.014      | -0.015    | 0.000    | 0.443 | 0.625   |
| obs coarse      | 463  | 0.029     | 0.017      | -0.012    | 0.000    | 0.457 | 0.604   |
| obs coarse      | 509  | 0.030     | 0.015      | -0.016    | 0.000    | 0.441 | 0.635   |
| obs coarse      | 571  | 0.029     | 0.014      | -0.015    | 0.000    | 0.464 | 0.625   |
| obs coarse      | 631  | 0.033     | 0.016      | -0.017    | 0.000    | 0.433 | 0.557   |
| obs coarse      | 701  | 0.026     | 0.015      | -0.012    | 0.000    | 0.477 | 0.656   |
| obs coarse      | 761  | 0.028     | 0.016      | -0.012    | 0.000    | 0.464 | 0.620   |
| obs coarse      | 823  | 0.032     | 0.016      | -0.016    | 0.000    | 0.445 | 0.573   |
| obs coarse      | 887  | 0.033     | 0.017      | -0.017    | 0.000    | 0.425 | 0.594   |
| obs coarse      | 947  | 0.032     | 0.014      | -0.018    | 0.000    | 0.454 | 0.609   |
| obs blind       | 11   | 0.029     | 0.028      | -0.001    | 0.000    | 0.442 | 0.568   |
| obs blind       | 29   | 0.029     | 0.025      | -0.004    | 0.000    | 0.444 | 0.630   |
| obs blind       | 47   | 0.025     | 0.023      | -0.002    | 0.000    | 0.454 | 0.651   |

| Case              | Seed | Group MSE | Critic MSE | Delta MSE | Zero-var | Wait  | Success |
|-------------------|------|-----------|------------|-----------|----------|-------|---------|
| obs blind         | 83   | 0.030     | 0.026      | -0.004    | 0.000    | 0.457 | 0.589   |
| obs blind         | 131  | 0.031     | 0.027      | -0.003    | 0.000    | 0.426 | 0.615   |
| obs blind         | 173  | 0.032     | 0.028      | -0.004    | 0.000    | 0.442 | 0.599   |
| obs blind         | 211  | 0.029     | 0.025      | -0.005    | 0.000    | 0.450 | 0.615   |
| obs blind         | 257  | 0.033     | 0.029      | -0.003    | 0.000    | 0.435 | 0.542   |
| obs blind         | 307  | 0.029     | 0.027      | -0.002    | 0.000    | 0.441 | 0.615   |
| obs blind         | 359  | 0.031     | 0.025      | -0.006    | 0.000    | 0.444 | 0.604   |
| obs blind         | 401  | 0.030     | 0.027      | -0.003    | 0.000    | 0.437 | 0.620   |
| obs blind         | 463  | 0.029     | 0.026      | -0.003    | 0.000    | 0.451 | 0.594   |
| obs blind         | 509  | 0.030     | 0.025      | -0.005    | 0.000    | 0.436 | 0.641   |
| obs blind         | 571  | 0.030     | 0.025      | -0.005    | 0.000    | 0.454 | 0.620   |
| obs blind         | 631  | 0.033     | 0.029      | -0.005    | 0.000    | 0.426 | 0.562   |
| obs blind         | 701  | 0.027     | 0.024      | -0.004    | 0.000    | 0.470 | 0.641   |
| obs blind         | 761  | 0.028     | 0.024      | -0.004    | 0.000    | 0.468 | 0.630   |
| obs blind         | 823  | 0.032     | 0.026      | -0.006    | 0.000    | 0.435 | 0.562   |
| obs blind         | 887  | 0.032     | 0.028      | -0.004    | 0.000    | 0.423 | 0.604   |
| obs blind         | 947  | 0.031     | 0.027      | -0.004    | 0.000    | 0.449 | 0.604   |
| blind undercoverd | 11   | 0.051     | 0.055      | 0.004     | 0.000    | 0.313 | 0.398   |
| blind undercoverd | 29   | 0.053     | 0.052      | -0.002    | 0.000    | 0.324 | 0.414   |
| blind undercoverd | 47   | 0.048     | 0.052      | 0.004     | 0.000    | 0.330 | 0.434   |
| blind undercoverd | 83   | 0.047     | 0.053      | 0.006     | 0.000    | 0.357 | 0.430   |
| blind undercoverd | 131  | 0.049     | 0.054      | 0.004     | 0.000    | 0.315 | 0.434   |
| blind undercoverd | 173  | 0.048     | 0.049      | 0.001     | 0.000    | 0.316 | 0.469   |
| blind undercoverd | 211  | 0.049     | 0.050      | 0.001     | 0.000    | 0.327 | 0.461   |
| blind undercoverd | 257  | 0.051     | 0.055      | 0.004     | 0.000    | 0.309 | 0.406   |
| blind undercoverd | 307  | 0.043     | 0.046      | 0.003     | 0.000    | 0.339 | 0.477   |
| blind undercoverd | 359  | 0.050     | 0.057      | 0.007     | 0.000    | 0.315 | 0.426   |
| blind undercoverd | 401  | 0.047     | 0.053      | 0.005     | 0.000    | 0.334 | 0.398   |
| blind undercoverd | 463  | 0.049     | 0.051      | 0.002     | 0.000    | 0.352 | 0.391   |
| blind undercoverd | 509  | 0.048     | 0.050      | 0.002     | 0.000    | 0.348 | 0.434   |
| blind undercoverd | 571  | 0.051     | 0.052      | 0.001     | 0.000    | 0.362 | 0.418   |
| blind undercoverd | 631  | 0.053     | 0.055      | 0.002     | 0.000    | 0.297 | 0.418   |
| blind undercoverd | 701  | 0.051     | 0.054      | 0.003     | 0.000    | 0.323 | 0.430   |
| blind undercoverd | 761  | 0.051     | 0.057      | 0.006     | 0.000    | 0.316 | 0.438   |
| blind undercoverd | 823  | 0.045     | 0.049      | 0.004     | 0.000    | 0.332 | 0.449   |
| blind undercoverd | 887  | 0.048     | 0.051      | 0.003     | 0.000    | 0.343 | 0.426   |
| blind undercoverd | 947  | 0.048     | 0.054      | 0.006     | 0.000    | 0.322 | 0.449   |
| sparse group04    | 11   | 0.021     | 0.007      | -0.014    | 0.000    | 0.407 | 0.486   |
| sparse group04    | 29   | 0.023     | 0.007      | -0.016    | 0.000    | 0.442 | 0.528   |
| sparse group04    | 47   | 0.020     | 0.007      | -0.013    | 0.000    | 0.413 | 0.472   |
| sparse group04    | 83   | 0.024     | 0.007      | -0.017    | 0.000    | 0.432 | 0.458   |
| sparse group04    | 131  | 0.019     | 0.006      | -0.013    | 0.000    | 0.433 | 0.479   |
| sparse group04    | 173  | 0.024     | 0.006      | -0.018    | 0.000    | 0.399 | 0.472   |
| sparse group04    | 211  | 0.026     | 0.008      | -0.018    | 0.000    | 0.404 | 0.493   |
| sparse group04    | 257  | 0.025     | 0.007      | -0.018    | 0.000    | 0.419 | 0.431   |
| sparse group04    | 307  | 0.025     | 0.007      | -0.018    | 0.000    | 0.412 | 0.403   |
| sparse group04    | 359  | 0.022     | 0.008      | -0.014    | 0.000    | 0.412 | 0.493   |
| sparse group04    | 401  | 0.023     | 0.008      | -0.014    | 0.000    | 0.417 | 0.479   |
| sparse group04    | 463  | 0.020     | 0.005      | -0.014    | 0.000    | 0.435 | 0.507   |
| sparse group04    | 509  | 0.024     | 0.007      | -0.017    | 0.000    | 0.439 | 0.424   |
| sparse group04    | 571  | 0.027     | 0.009      | -0.018    | 0.000    | 0.408 | 0.472   |
| sparse group04    | 631  | 0.022     | 0.007      | -0.015    | 0.000    | 0.399 | 0.451   |
| sparse group04    | 701  | 0.023     | 0.007      | -0.015    | 0.000    | 0.435 | 0.472   |
| sparse group04    | 761  | 0.022     | 0.009      | -0.013    | 0.000    | 0.429 | 0.493   |
| sparse group04    | 823  | 0.026     | 0.005      | -0.021    | 0.000    | 0.410 | 0.500   |
| sparse group04    | 887  | 0.025     | 0.007      | -0.018    | 0.000    | 0.412 | 0.479   |
| sparse group04    | 947  | 0.022     | 0.008      | -0.015    | 0.000    | 0.385 | 0.500   |
| sparse group08    | 11   | 0.023     | 0.004      | -0.019    | 0.000    | 0.409 | 0.497   |
| sparse group08    | 29   | 0.024     | 0.005      | -0.019    | 0.000    | 0.434 | 0.455   |
| sparse group08    | 47   | 0.021     | 0.005      | -0.016    | 0.000    | 0.425 | 0.462   |
| sparse group08    | 83   | 0.022     | 0.004      | -0.018    | 0.000    | 0.417 | 0.455   |
| sparse group08    | 131  | 0.022     | 0.003      | -0.019    | 0.000    | 0.430 | 0.458   |
| sparse group08    | 173  | 0.022     | 0.004      | -0.018    | 0.000    | 0.415 | 0.469   |
| sparse group08    | 211  | 0.024     | 0.004      | -0.020    | 0.000    | 0.405 | 0.455   |
| sparse group08    | 257  | 0.023     | 0.005      | -0.018    | 0.000    | 0.414 | 0.462   |
| sparse group08    | 307  | 0.023     | 0.004      | -0.019    | 0.000    | 0.419 | 0.417   |

| Case           | Seed | Group MSE | Critic MSE | Delta MSE | Zero-var | Wait  | Success |
|----------------|------|-----------|------------|-----------|----------|-------|---------|
| sparse group08 | 359  | 0.022     | 0.005      | -0.017    | 0.000    | 0.419 | 0.472   |
| sparse group08 | 401  | 0.022     | 0.005      | -0.017    | 0.000    | 0.417 | 0.479   |
| sparse group08 | 463  | 0.022     | 0.003      | -0.019    | 0.000    | 0.429 | 0.507   |
| sparse group08 | 509  | 0.023     | 0.004      | -0.019    | 0.000    | 0.428 | 0.476   |
| sparse group08 | 571  | 0.024     | 0.004      | -0.020    | 0.000    | 0.415 | 0.427   |
| sparse group08 | 631  | 0.021     | 0.004      | -0.017    | 0.000    | 0.417 | 0.465   |
| sparse group08 | 701  | 0.021     | 0.004      | -0.017    | 0.000    | 0.434 | 0.490   |
| sparse group08 | 761  | 0.023     | 0.004      | -0.019    | 0.000    | 0.434 | 0.479   |
| sparse group08 | 823  | 0.024     | 0.005      | -0.019    | 0.000    | 0.402 | 0.444   |
| sparse group08 | 887  | 0.025     | 0.004      | -0.021    | 0.000    | 0.419 | 0.476   |
| sparse group08 | 947  | 0.020     | 0.004      | -0.017    | 0.000    | 0.416 | 0.465   |

## G Reproducibility Checklist

1. Run the unit tests with `python3 -m unittest discover -s tests`.
2. Run the smoke gate with `./scripts/run_smoke.sh`.
3. Regenerate the deep matrix with `python3 -m experiments.deep_matrix`.
4. Regenerate LaTeX inputs with `python3 scripts/build_latex_inputs.py`.
5. Build the full paper with `./scripts/build_latex_paper.sh`.
6. Verify that the LaTeX artifact manifest records the PDF hash and page count.
7. Treat `public/ppo_grpo_opd_long_horizon.pdf` and the DOCX as abridged reports; treat the LaTeX PDF as the full paper artifact.

## H Next Experiments

**Closed-loop toy training.** The next toy should train parameterized policies with a shared reward and compare closed-loop PPO, GRPO, and corrected GRPO variants under equal generated-token budgets. This will test whether estimator fidelity predicts policy improvement.

**Process rewards.** Adding process labels or subgoal rewards should narrow the gap between critic-based and group-relative estimators. This prediction is important because many practical systems do not rely on terminal reward alone.

**Compaction.** A compaction experiment should split long traces into variable numbers of sub-traces. The hypothesis is that critic-based individual-rollout objectives handle this naturally, while group-wise objectives need additional rules for matching sub-traces across completions.

**Distillation hybrids.** An OPD/OPSD follow-up should use successful RL traces as teacher or privileged self-teacher material and measure whether dense distillation preserves the credit assignment gains at lower inference cost.

## I Replication Protocol for New Benchmarks

**Trace schema.** A real long-horizon benchmark should record more than a final reward. Each trajectory should include prompt identity, sampled response identity, token or step indices, tool calls, tool observations, compaction boundaries, verifier calls, process labels when available, terminal reward, and any safety or refusal events. This schema is the minimum needed to compare a group-relative estimator with a value-model estimator after the fact. Without it, the experiment can still report final success, but it cannot diagnose where credit was assigned or lost.

**Budget matching.** The clean comparison is not “PPO run” versus “GRPO run” in isolation. The comparison should state generated-token budget, number of prompts, completions per prompt, verifier calls, critic update steps, policy update steps, and peak activation or optimizer memory. A critic-based method should be charged for critic fitting and storage. A group-relative method should be charged for the extra completions needed to obtain reward contrast. A distillation method should be charged for teacher inference or self-teacher selection.

**Primary measurements.** For any benchmark with oracle or process annotations, report estimator correlation with oracle credit, calibrated MSE, sign accuracy, zero-variance group fraction, and return improvement under equal rollout budgets. For benchmarks without oracle credit, report held-out process-label agreement, per-step value calibration where labels exist, and sensitivity to trajectory length. Final score alone is not enough because a method can improve a short task while failing the mechanism required for long traces.

**Decision rule.** A public claim that one family is better should require at least three checks: the mean effect favors the method, the confidence interval excludes practical near-ties, and an adversarial or counterexample regime is reported. This paper’s toy result follows that rule: it highlights the critic-favorable mean pattern, the near tie under tiny critic budget, and the group-favorable blind undercovered case. A larger benchmark should preserve the same discipline.

**Audit package.** The public release should include raw per-seed rows, aggregate tables, plotting scripts, exact command lines, environment notes, and a short statement of non-claims. The non-claims matter. For this workstream, the honest claim is that critic-style temporal information can dominate in observable long traces, while group-relative normalization remains strong when reward contrast is reliable and value modeling is weak. That is a research program, not a slogan.

## J Artifact Provenance and Public Release Notes

**Canonical data source.** The canonical numeric artifact for this paper is the deep-matrix JSON under `results/`. The LaTeX macros, selected result table, axis summary, full case summary, raw seed rows, and calibrated-error appendix are generated from that JSON file by the LaTeX input builder. This keeps the manuscript from silently drifting away from the experiment output.

**Rendered artifacts.** The repository intentionally contains two kinds of public artifacts. The abridged PDF/DOCX report is designed for quick reading and sharing. The LaTeX PDF is the full paper artifact, including appendices and raw tables. The two artifacts should not be compared by page count; they serve different review contexts.

**Manifest interpretation.** The LaTeX manifest records input hashes, output hashes, the TeX engine, and the page-count gate. Because generated artifacts are committed with their manifest, the manifest’s recorded git status necessarily describes the generation-time worktree rather than the final commit that contains the manifest. A clean checkout can rerun the build script to refresh the manifest and should recover the same result counts from the canonical JSON.

**How to read the claim.** The public headline should be the cautious one: 16 clear critic-favorable cases, 1 near tie, and 1 clear group-favorable counterexample in this toy estimator benchmark. The stronger-sounding “17 mean critic wins” is true as a descriptive statistic, but the near-tie is scientifically important. The counterexample is equally important because it states a boundary condition for GRPO-style methods.

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